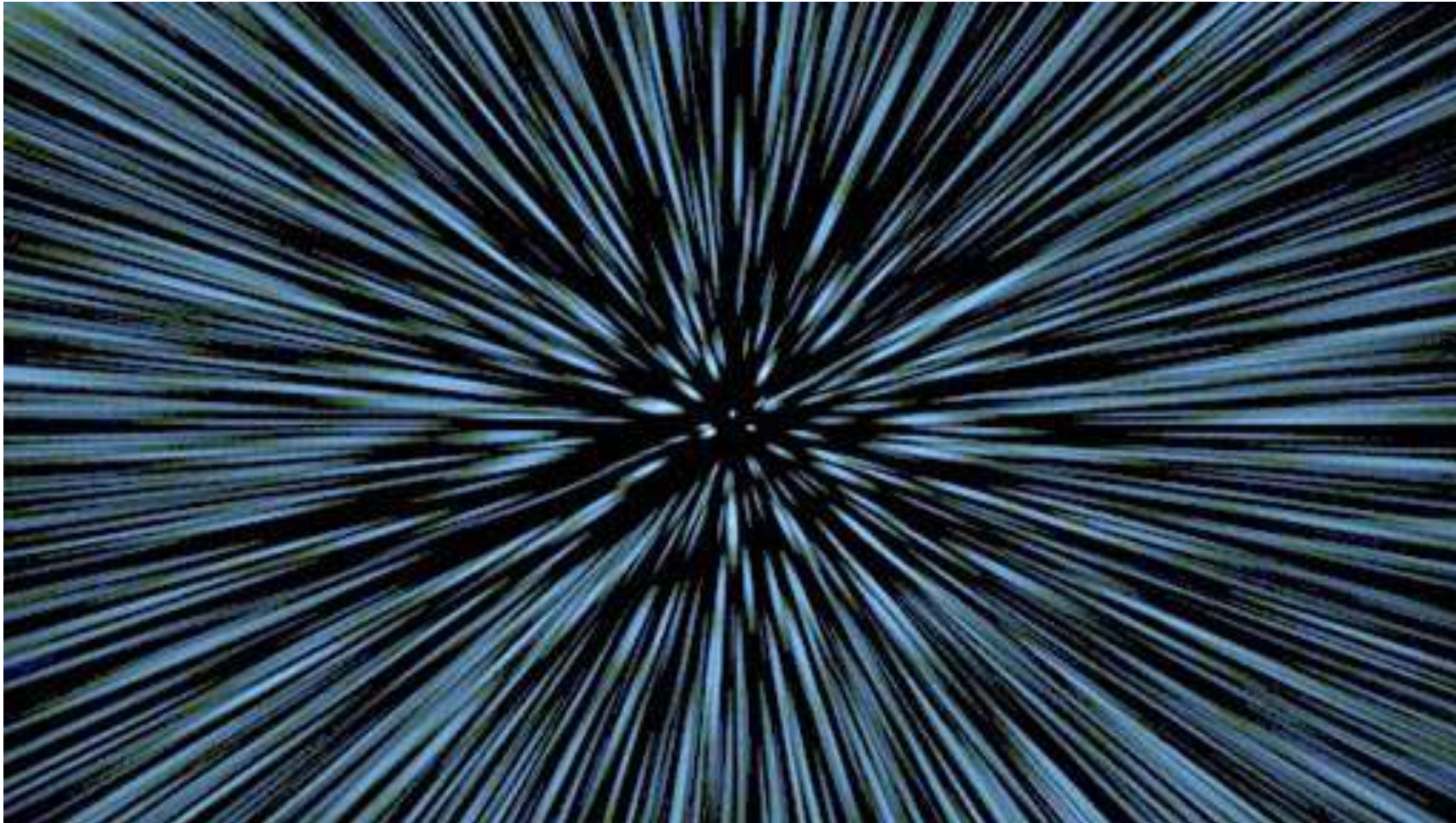


2D transformations (a.k.a. warping)



Reminder: image transformations

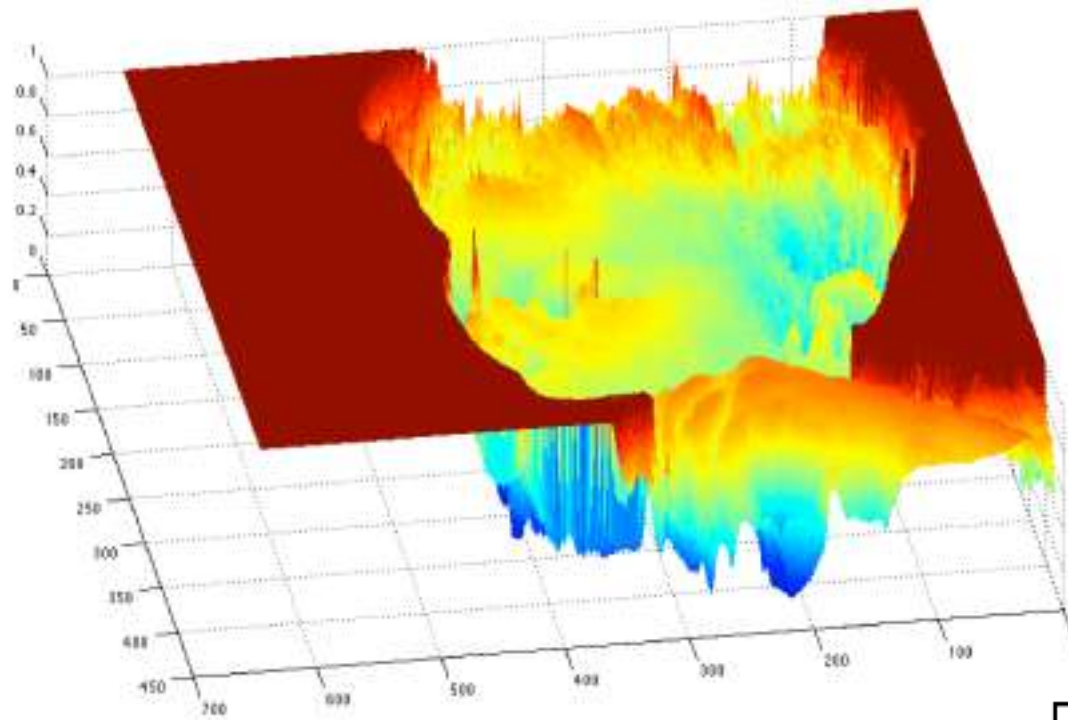
What is an image?

$$f(\mathbf{x})$$



grayscale image

What is the range of the image function f ?



domain $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$

A (grayscale) image is a 2D function.

What types of image transformations can we do?



Filtering



changes pixel *values*



Warping



changes pixel *locations*

What types of image transformations can we do?

F



Filtering



$$G(\mathbf{x}) = h\{F(\mathbf{x})\}$$

G



changes *range* of image function

F



Warping



$$G(\mathbf{x}) = F(h\{\mathbf{x}\})$$

G



changes *domain* of image function

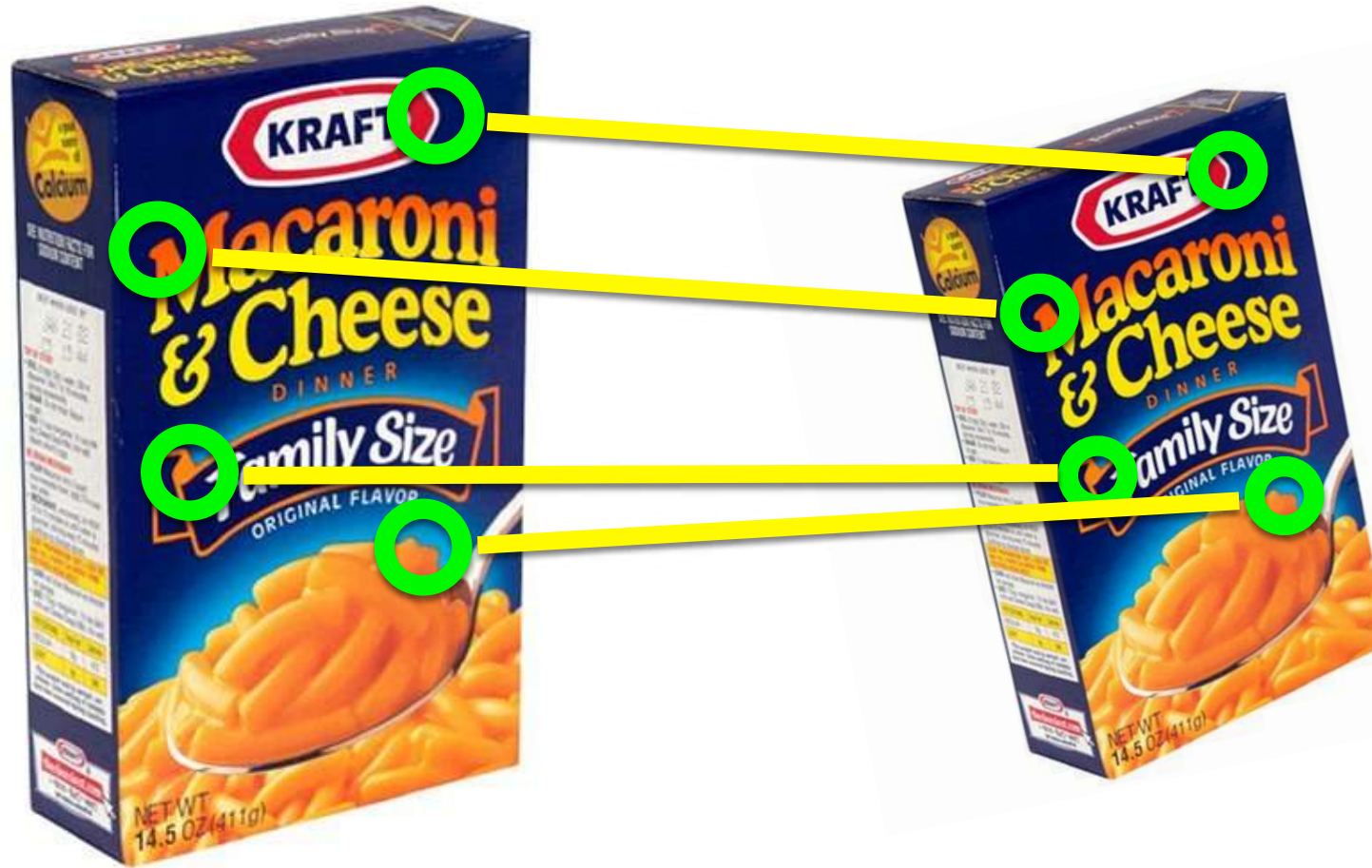
Warping example: feature matching



Warping example: feature matching



Warping example: feature matching

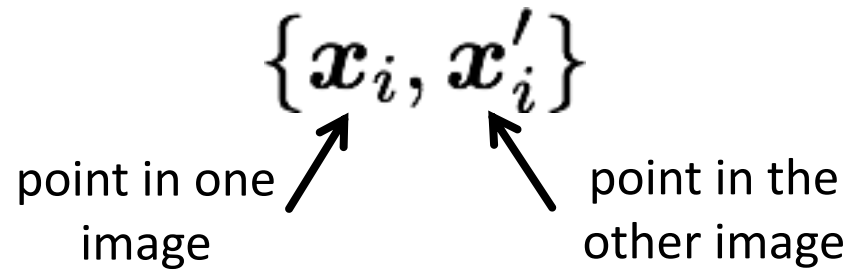


- object recognition
- 3D reconstruction
- augmented reality
- image stitching

How do you compute the transformation?

Warping example: feature matching

Given a set of matched feature points:



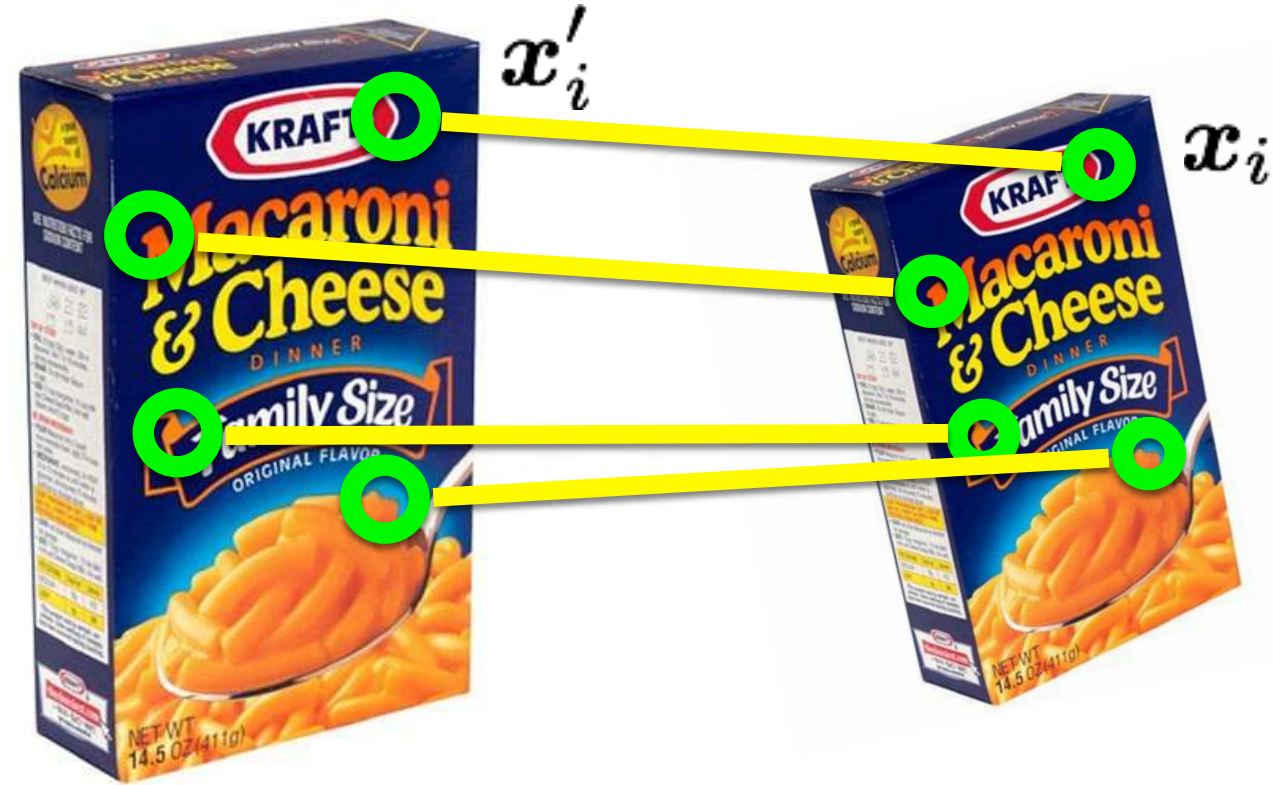
and a transformation:

$$x' = f(x; p)$$

transformation function parameters

find the best estimate of the parameters

p



What kind of transformation functions f are there?

2D transformations

2D transformations



translation



rotation



aspect



affine



perspective



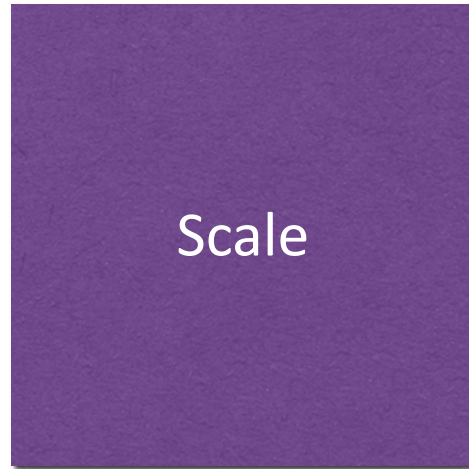
cylindrical

2D planar transformations



2D planar transformations

y



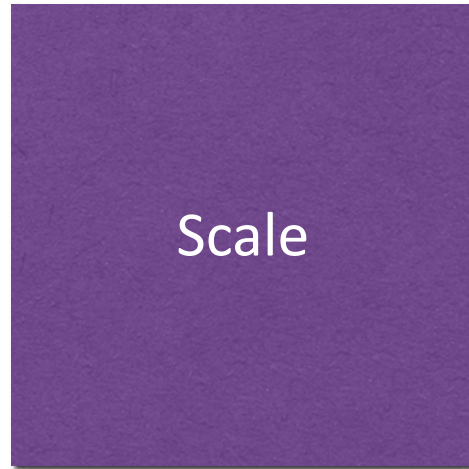
How would you implement scaling?

- Each component multiplied by a scalar
- Uniform scaling - same scalar for each component

x

2D planar transformations

y



$$x' = ax$$

$$y' = by$$

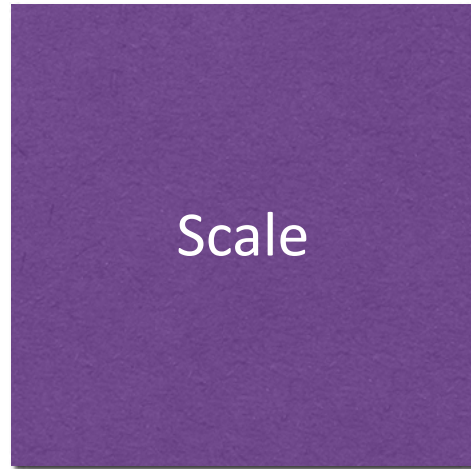
What's the effect of using
different scale factors?

- Each component multiplied by a scalar
- Uniform scaling - same scalar for each component

x

2D planar transformations

y



$$x' = ax$$

$$y' = by$$

matrix representation of scaling:

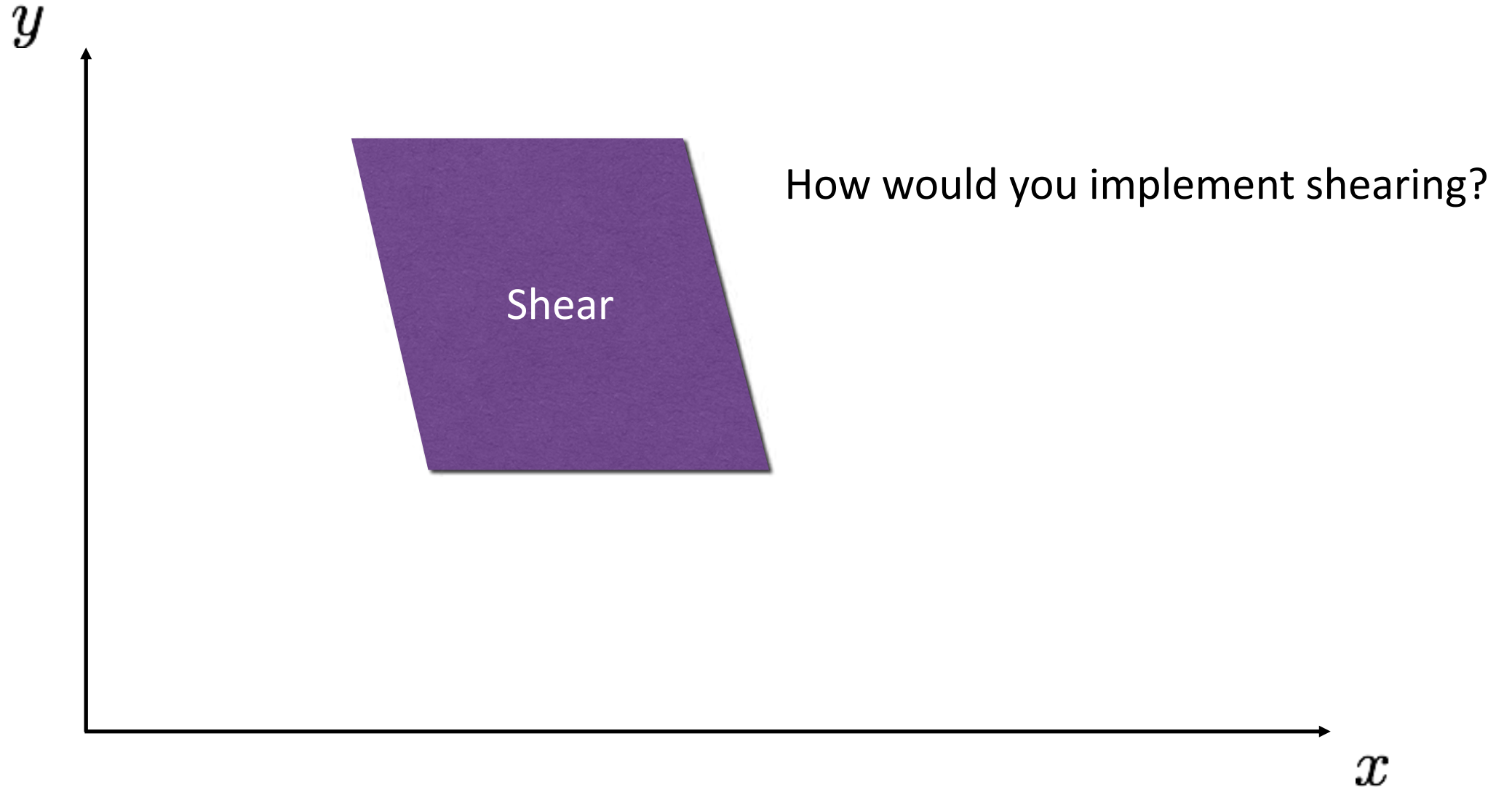
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \underbrace{\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}}_{\text{scaling matrix } S} \begin{bmatrix} x \\ y \end{bmatrix}$$

scaling matrix S

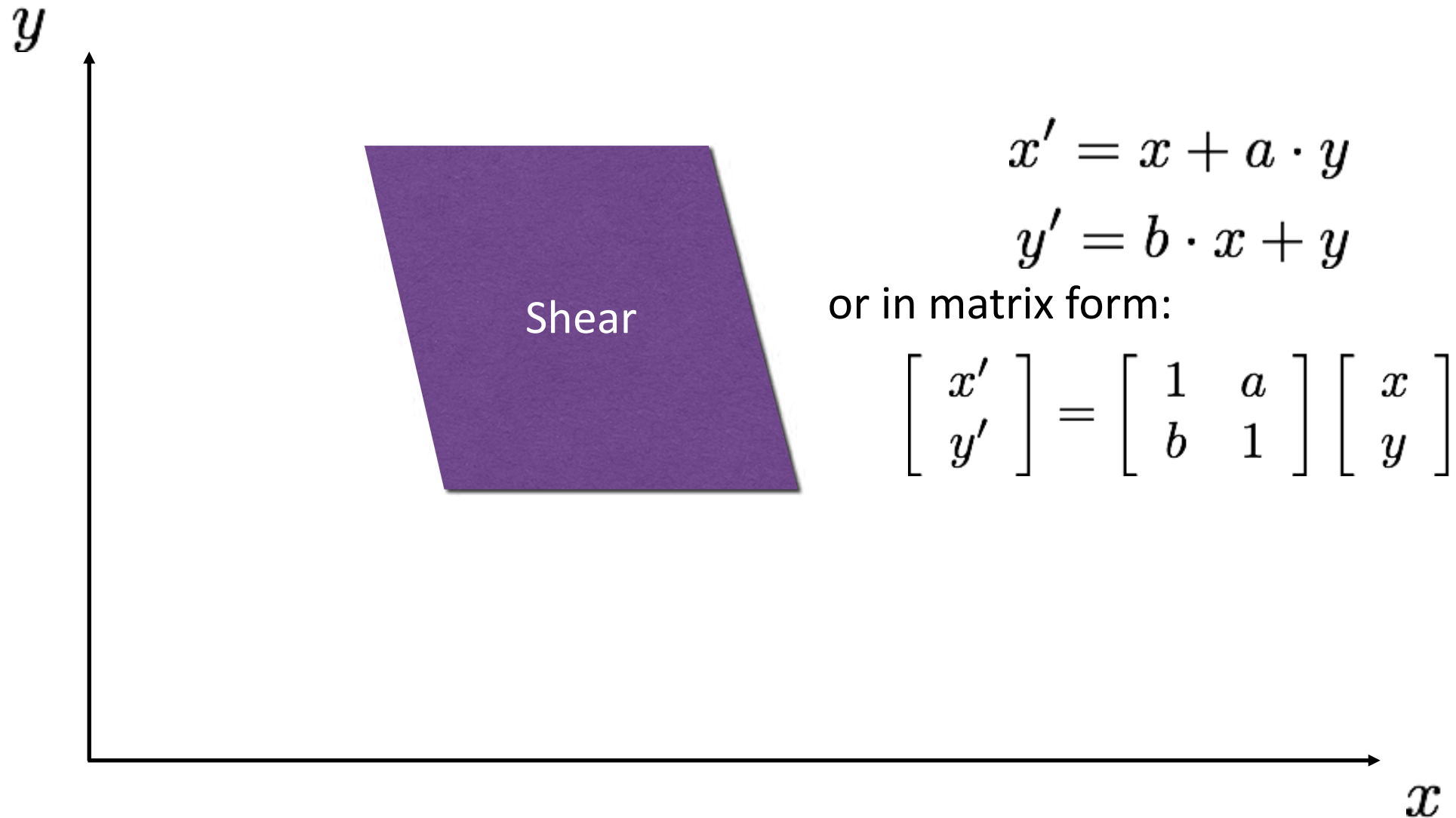
- Each component multiplied by a scalar
- Uniform scaling - same scalar for each component

x

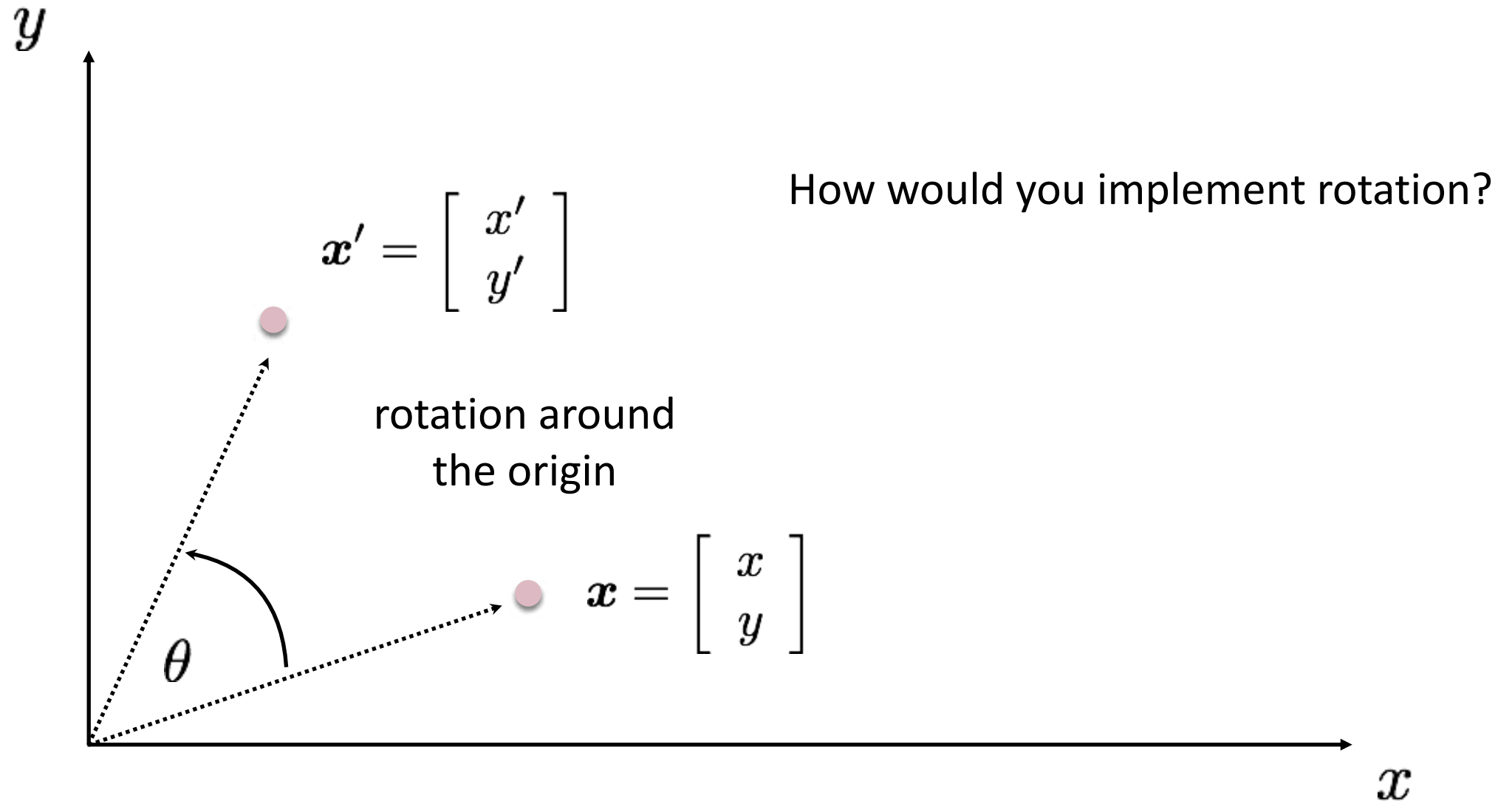
2D planar transformations



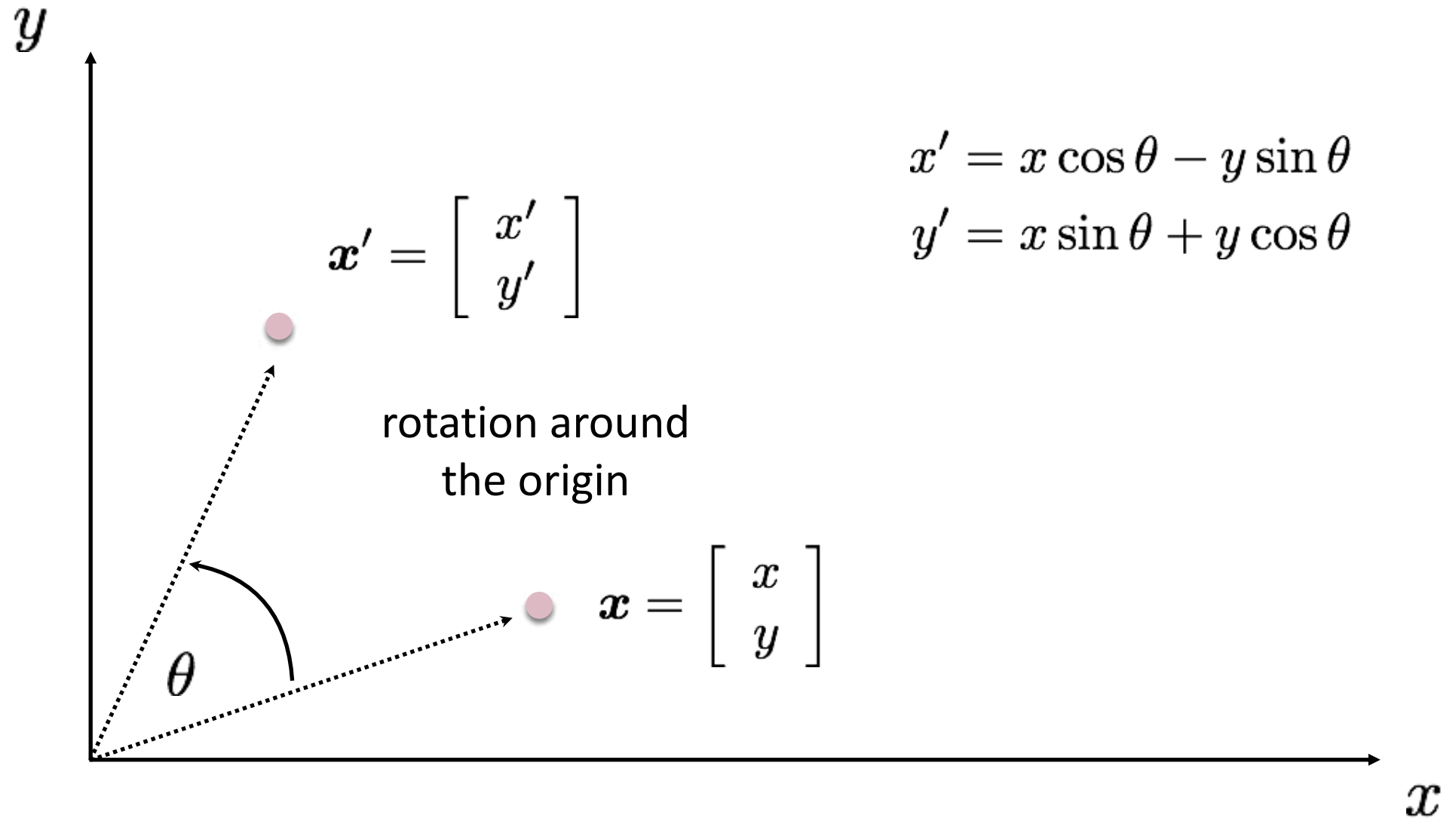
2D planar transformations



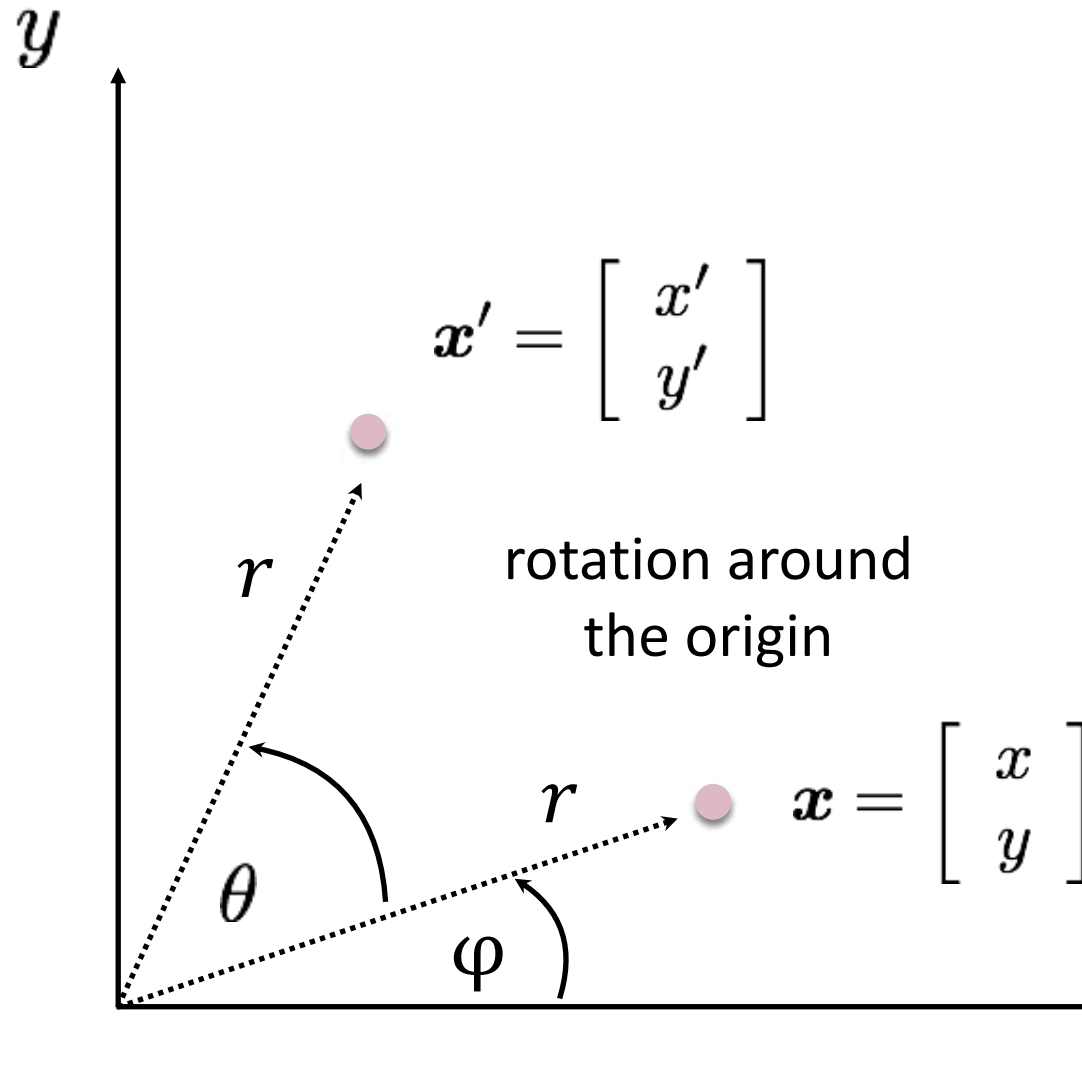
2D planar transformations



2D planar transformations



2D planar transformations



Polar coordinates...

$$x = r \cos(\phi)$$

$$y = r \sin(\phi)$$

$$x' = r \cos(\phi + \theta)$$

$$y' = r \sin(\phi + \theta)$$

Trigonometric Identity...

$$x' = r \cos(\phi) \cos(\theta) - r \sin(\phi) \sin(\theta)$$

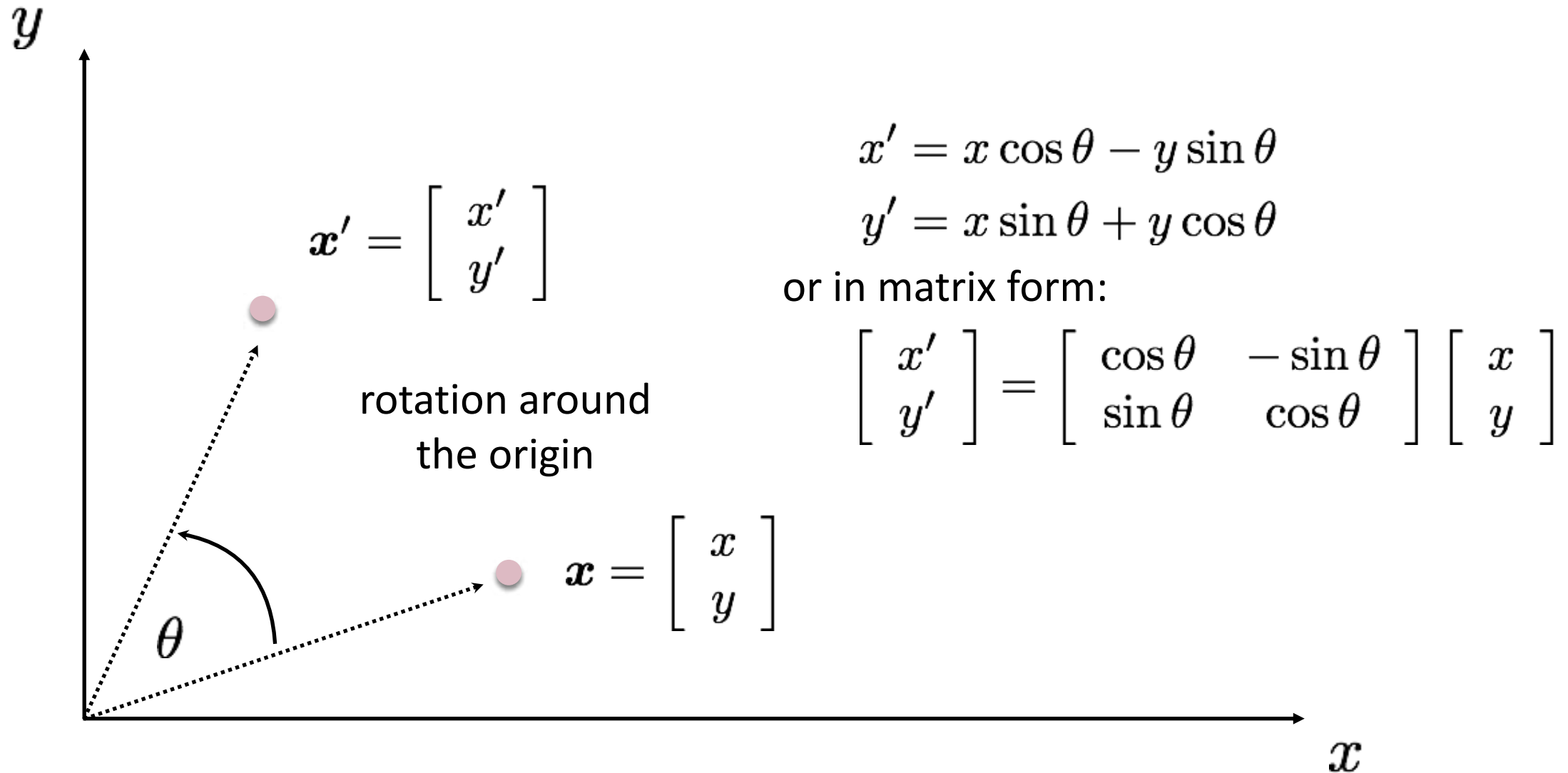
$$y' = r \sin(\phi) \cos(\theta) + r \cos(\phi) \sin(\theta)$$

Substitute...

$$x' = x \cos(\theta) - y \sin(\theta)$$

$$y' = x \sin(\theta) + y \cos(\theta)$$

2D planar transformations



2D planar and linear transformations

$$\boldsymbol{x}' = f(\boldsymbol{x}; p)$$



$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \boldsymbol{M} \begin{bmatrix} x \\ y \end{bmatrix}$$

parameters p

point $\boldsymbol{x}$

2D planar and linear transformations

Scale

$$\mathbf{M} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

Flip across y

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

Rotate

$$\mathbf{M} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Flip across origin

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

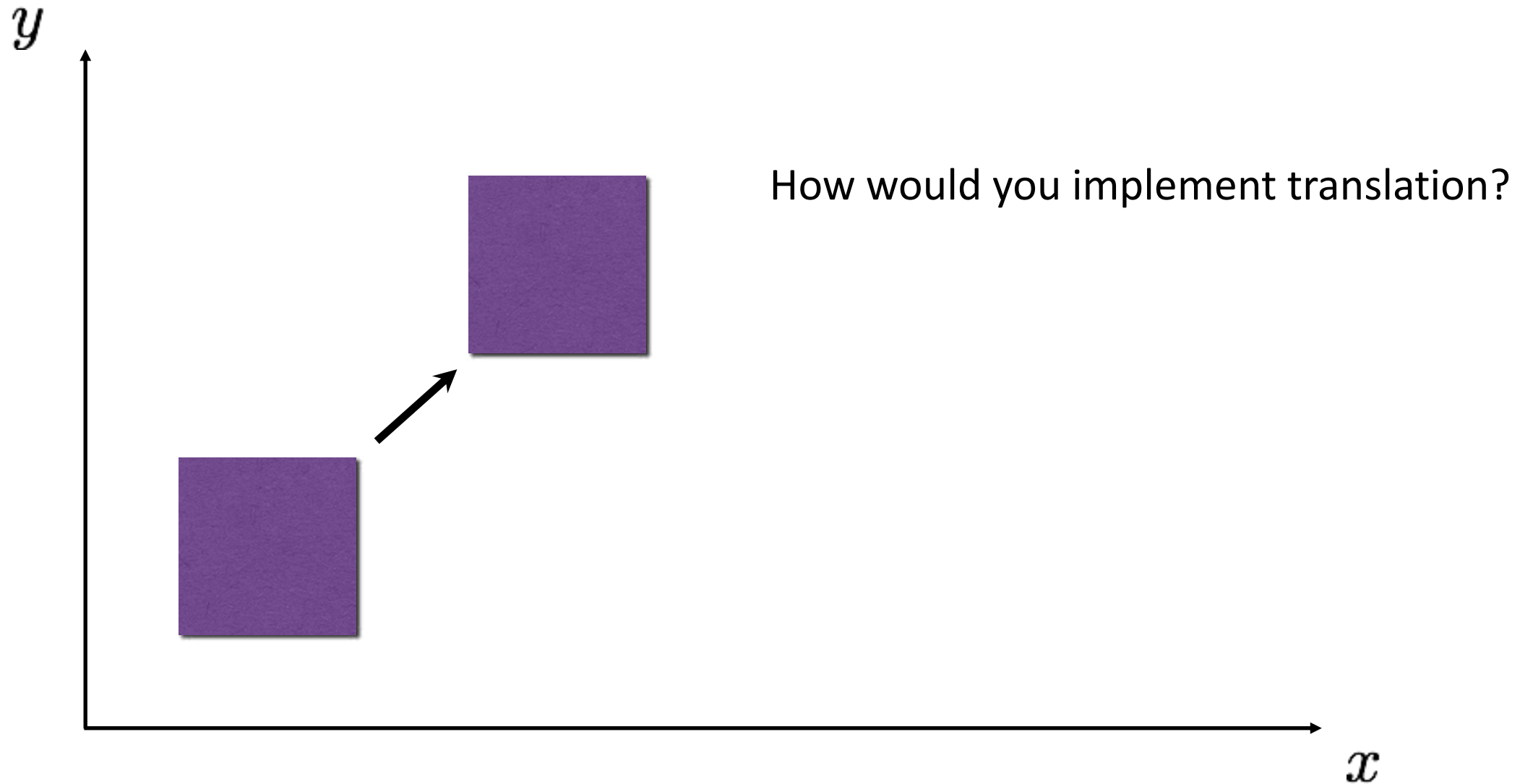
Shear

$$\mathbf{M} = \begin{bmatrix} 1 & s_x \\ s_y & 1 \end{bmatrix}$$

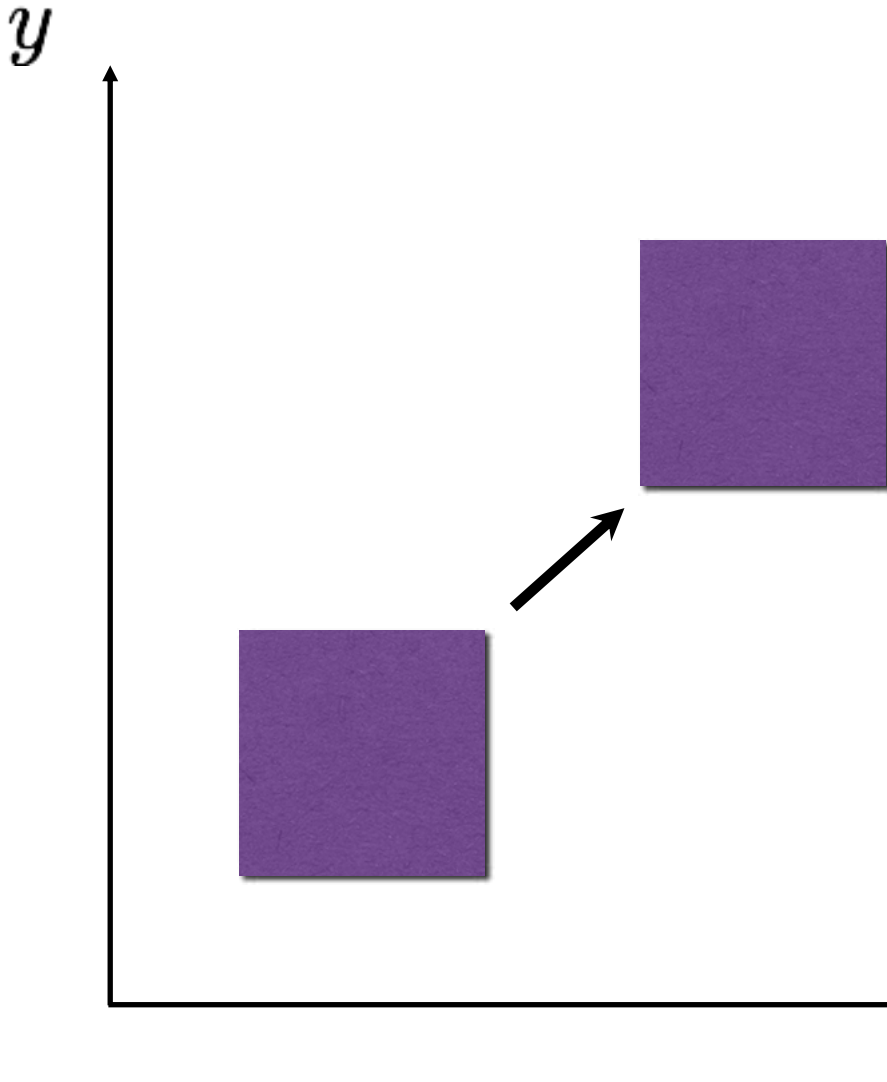
Identity

$$\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2D translation



2D translation



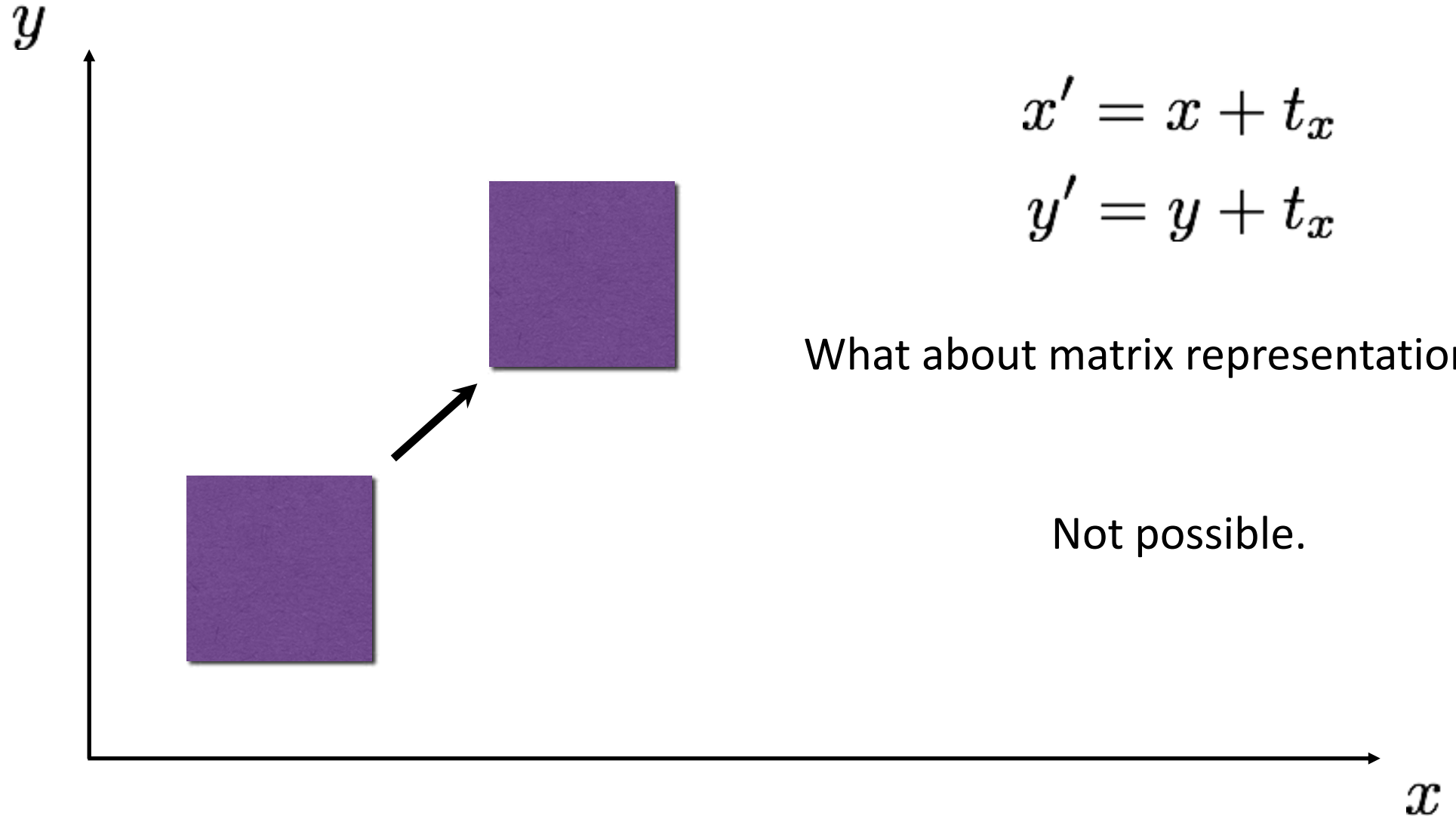
$$x' = x + t_x$$

$$y' = y + t_y$$

What about matrix representation?

$$\mathbf{M} = \begin{bmatrix} ? & ? \\ ? & ? \end{bmatrix}$$

2D translation



Projective geometry 101

Homogeneous coordinates

heterogeneous
coordinates

homogeneous
coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

add a 1 here



- Represent 2D point with a 3D vector

Homogeneous coordinates

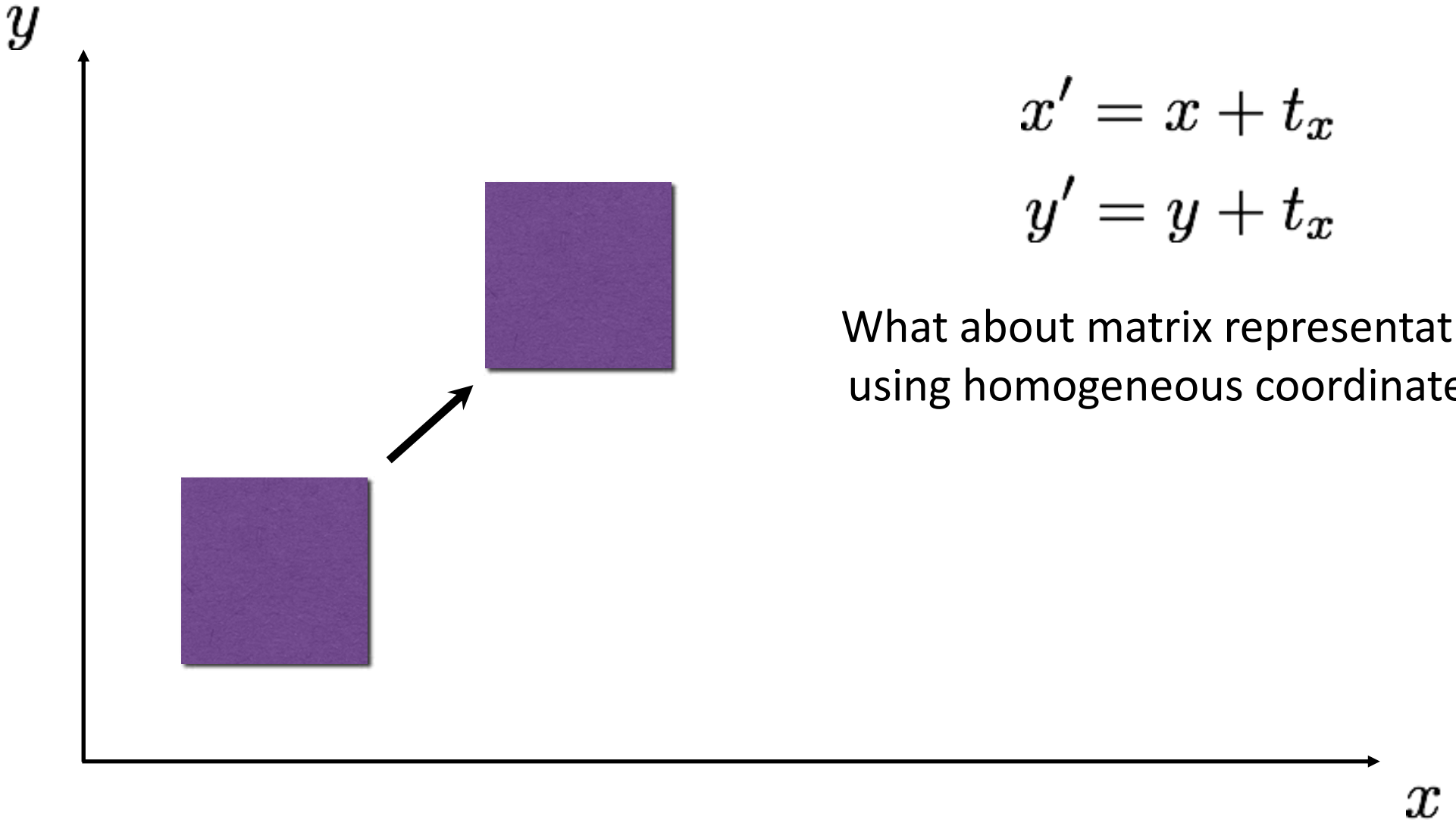
heterogeneous
coordinates

homogeneous
coordinates

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \stackrel{\text{def}}{=} \begin{bmatrix} ax \\ ay \\ a \end{bmatrix}$$

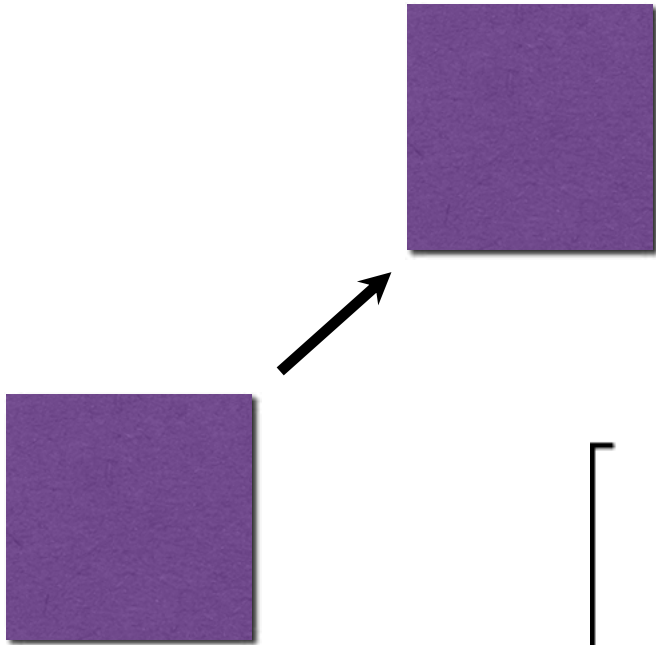
- Represent 2D point with a 3D vector
- 3D vectors are only defined up to scale

2D translation



2D translation

y



$$x' = x + t_x$$

$$y' = y + t_y$$

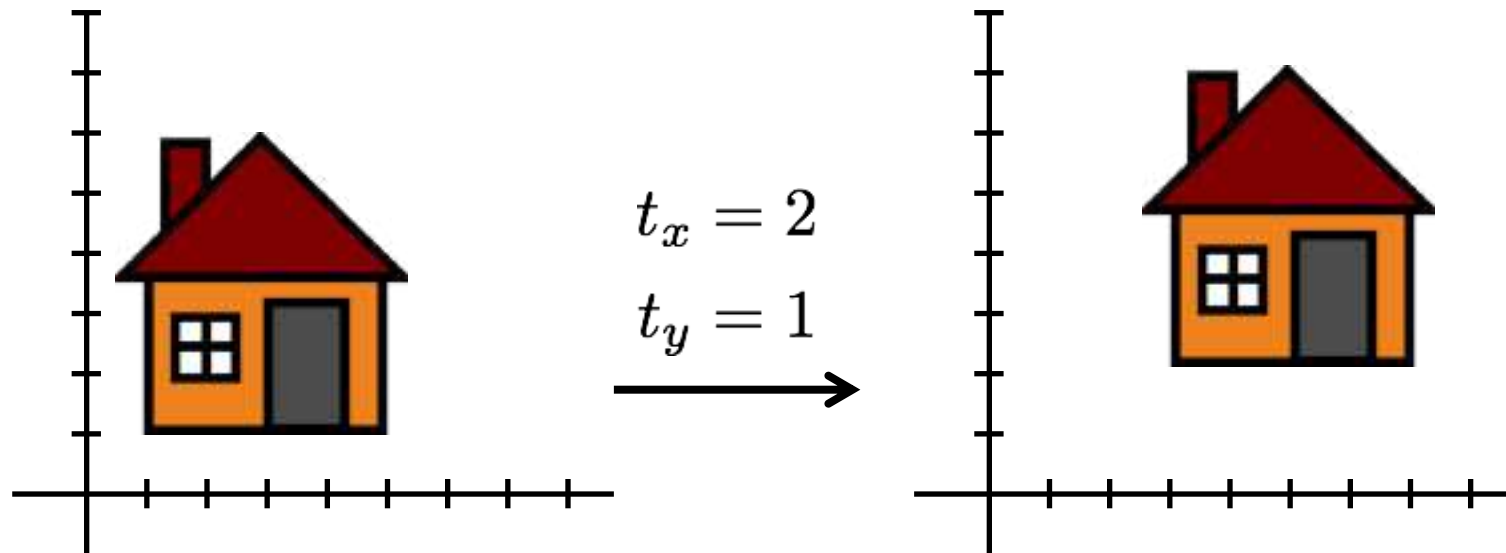
What about matrix representation
using homogeneous coordinates?

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \mathbf{M} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

x

2D translation using homogeneous coordinates

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + t_x \\ y + t_y \\ 1 \end{bmatrix}$$



Homogeneous coordinates

Conversion:

- heterogeneous $\rightarrow$ homogeneous

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- homogeneous $\rightarrow$ heterogeneous

$$\begin{bmatrix} x \\ y \\ w \end{bmatrix} \Rightarrow \begin{bmatrix} x/w \\ y/w \end{bmatrix}$$

- scale invariance

$$\begin{bmatrix} x & y & w \end{bmatrix}^{\top} = \lambda \begin{bmatrix} x & y & w \end{bmatrix}^{\top}$$

Special points:

- point at infinity

$$\begin{bmatrix} x & y & 0 \end{bmatrix}$$

- undefined

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

Projective geometry

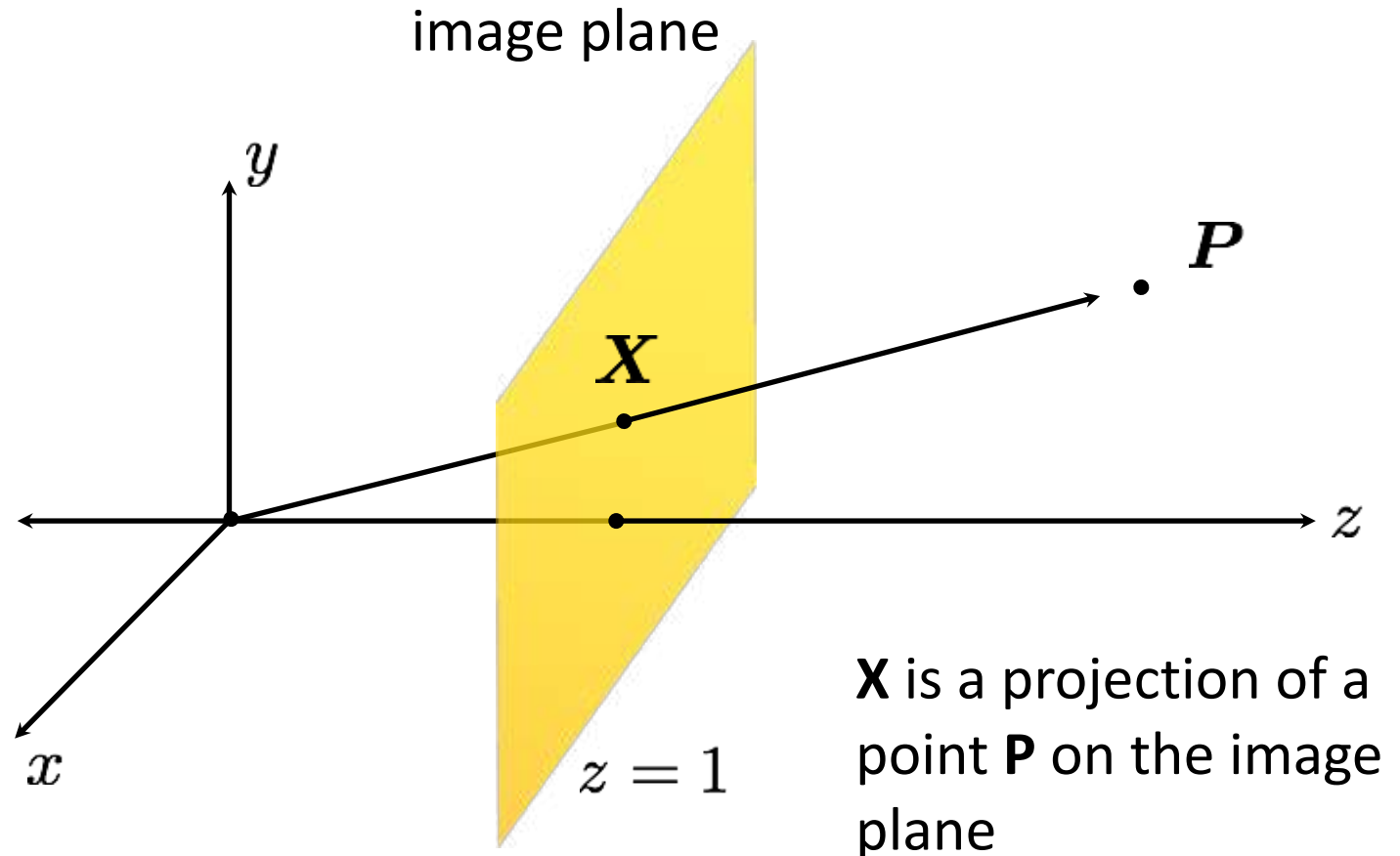
image point in
pixel coordinates

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$



image point in
homogeneous
coordinates

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Transformations in projective geometry

2D transformations in heterogeneous coordinates

Re-write these transformations as 3x3 matrices:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

shearing

2D transformations in heterogeneous coordinates

Re-write these transformations as 3x3 matrices:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

shearing

2D transformations in heterogeneous coordinates

Re-write these transformations as 3x3 matrices:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} & & \\ & ? & \\ & & \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta_x & 0 \\ \beta_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

shearing

2D transformations in heterogeneous coordinates

Re-write these transformations as 3x3 matrices:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

scaling

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & \beta_x & 0 \\ \beta_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

shearing

Matrix composition

Transformations can be combined by matrix multiplication:

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \left(\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

p' = ? ? ? **p**

Matrix composition

Transformations can be combined by matrix multiplication:

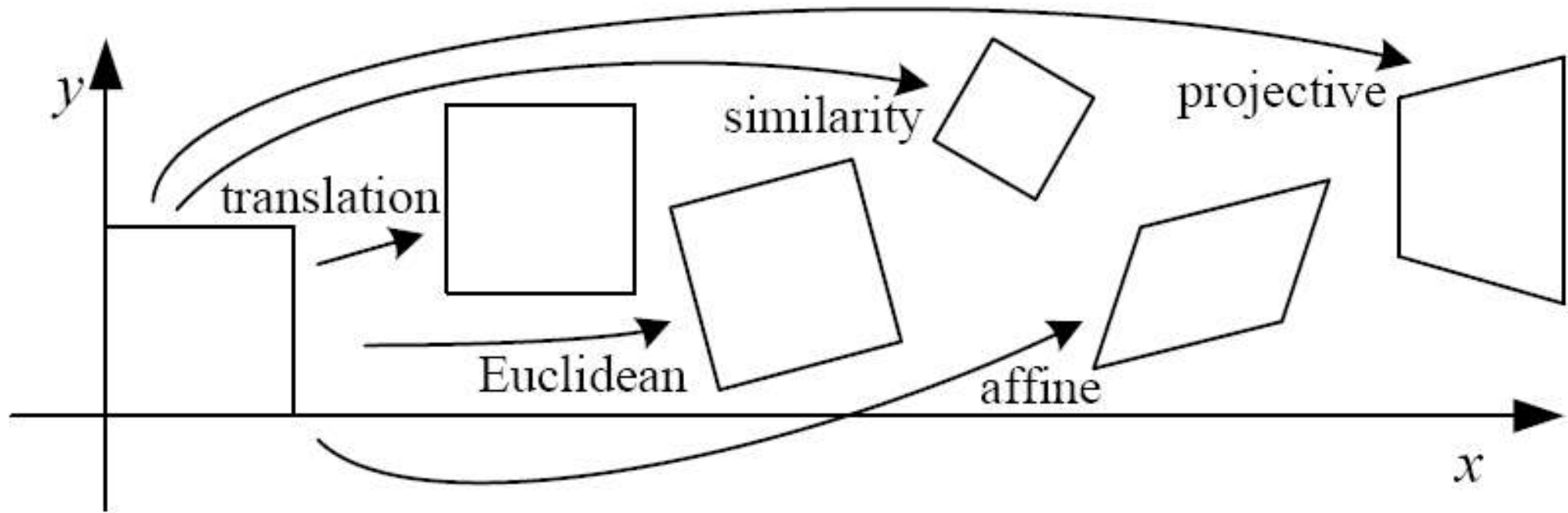
$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \left(\begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \Theta & -\sin \Theta & 0 \\ \sin \Theta & \cos \Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

$$\mathbf{p}' = \text{translation}(t_x, t_y) \quad \text{rotation}(\theta) \quad \text{scale}(s, s) \quad \mathbf{p}$$

Does the multiplication order matter?

Classification of 2D transformations

Classification of 2D transformations



Classification of 2D transformations

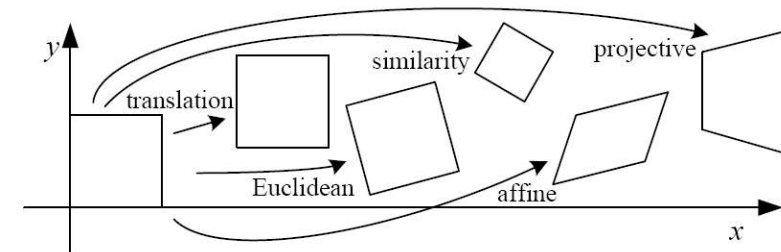
Name	Matrix	# D.O.F.
translation	$\begin{bmatrix} \mathbf{I} & & \mathbf{t} \end{bmatrix}$	?
rigid (Euclidean)	$\begin{bmatrix} \mathbf{R} & & \mathbf{t} \end{bmatrix}$	?
similarity	$\begin{bmatrix} s\mathbf{R} & & \mathbf{t} \end{bmatrix}$	?
affine	$\begin{bmatrix} \mathbf{A} \end{bmatrix}$	?
projective	$\begin{bmatrix} \tilde{\mathbf{H}} \end{bmatrix}$	?

Classification of 2D transformations

Translation:

$$\begin{bmatrix} 1 & 0 & t_1 \\ 0 & 1 & t_2 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?

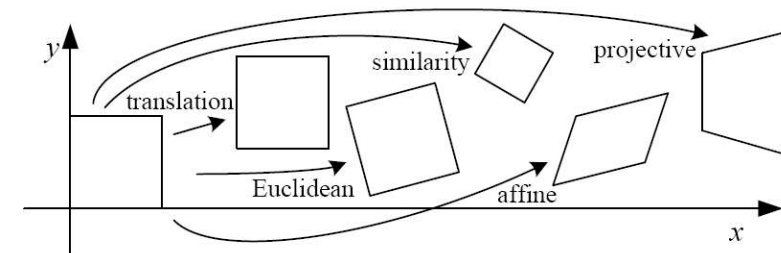


Classification of 2D transformations

Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

Are there any values that are related?

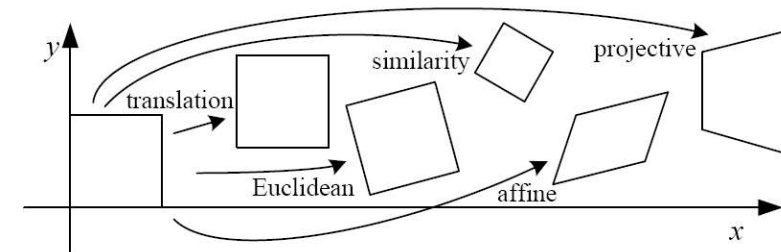


Classification of 2D transformations

Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?

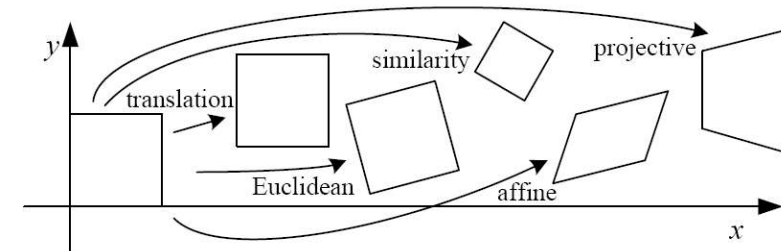


Classification of 2D transformations

which other matrix values
will change if this increases?

Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$



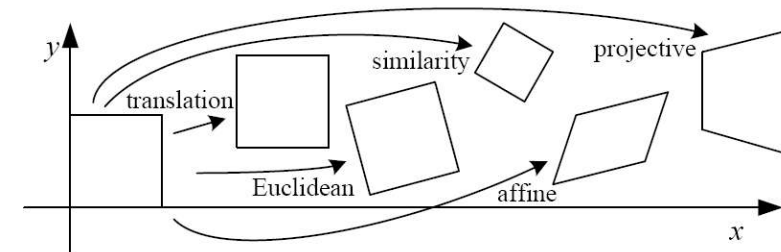
Classification of 2D transformations

what will happen to the
image if this increases?



Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$



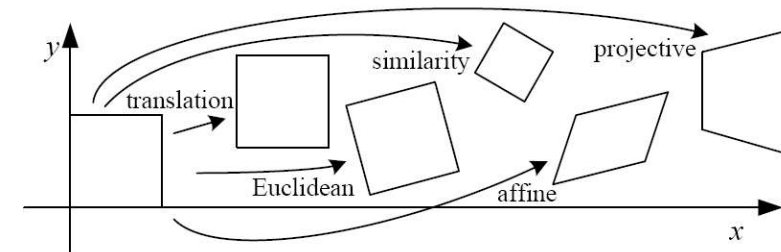
Classification of 2D transformations

what will happen to the
image if this increases?



Euclidean (rigid):
rotation + translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

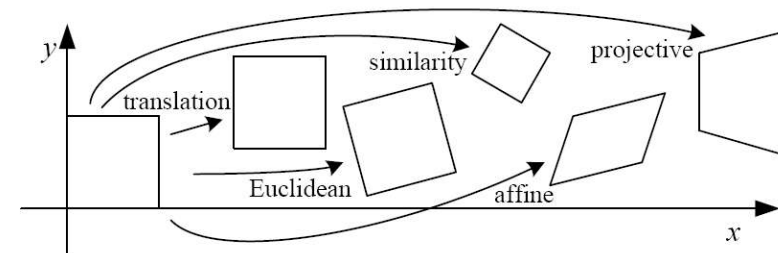


Classification of 2D transformations

Similarity:
uniform scaling + rotation
+ translation

$$\begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

Are there any values that are related?



Classification of 2D transformations

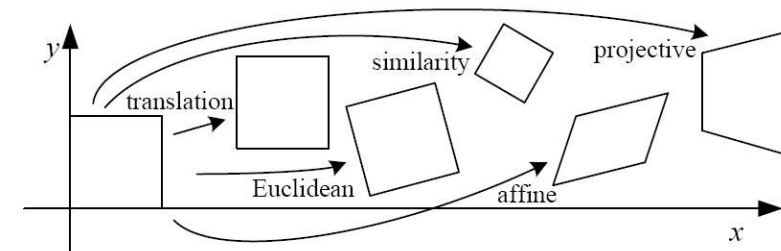
multiply these four by scale s



Similarity:
uniform scaling + rotation
+ translation

$$\begin{bmatrix} \cos \theta & -\sin \theta & r_3 \\ \sin \theta & \cos \theta & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?

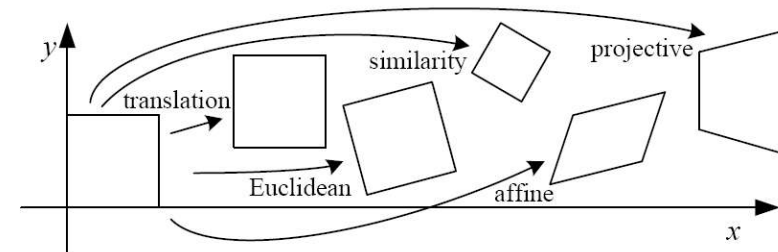


Classification of 2D transformations

what will happen to the
image if this increases?

Similarity:
uniform scaling + rotation
+ translation

$$\downarrow \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ 0 & 0 & 1 \end{bmatrix}$$

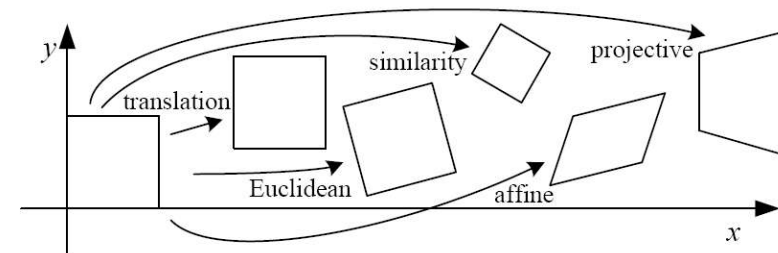


Classification of 2D transformations

Affine transform:
uniform scaling + shearing
+ rotation + translation

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ 0 & 0 & 1 \end{bmatrix}$$

Are there any values that are related?



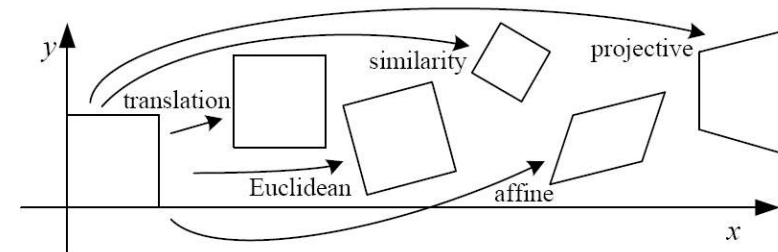
Classification of 2D transformations

Affine transform:
uniform scaling + shearing
+ rotation + translation

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ 0 & 0 & 1 \end{bmatrix}$$

Are there any values that are related?

$$\begin{array}{cc} \text{similarity} & \text{shear} \\ \begin{bmatrix} sr_1 & sr_2 \\ sr_3 & sr_4 \end{bmatrix} & \begin{bmatrix} 1 & h_1 \\ h_2 & 1 \end{bmatrix} \end{array} = \begin{bmatrix} sr_1 + h_2 sr_2 & sr_2 + h_1 sr_1 \\ sr_3 + h_2 sr_4 & sr_4 + h_1 sr_3 \end{bmatrix}$$



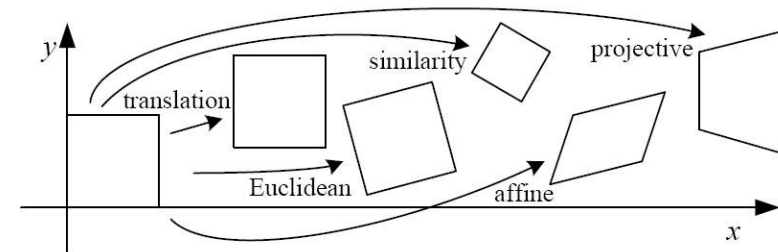
Classification of 2D transformations

Affine transform:
uniform scaling + shearing
+ rotation + translation

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ 0 & 0 & 1 \end{bmatrix}$$

How many degrees of freedom?

$$\begin{array}{cc} \text{similarity} & \text{shear} \\ \begin{bmatrix} sr_1 & sr_2 \\ sr_3 & sr_4 \end{bmatrix} & \begin{bmatrix} 1 & h_1 \\ h_2 & 1 \end{bmatrix} \end{array} = \begin{bmatrix} sr_1 + h_2 sr_2 & sr_2 + h_1 sr_1 \\ sr_3 + h_2 sr_4 & sr_4 + h_1 sr_3 \end{bmatrix}$$



Affine transformations

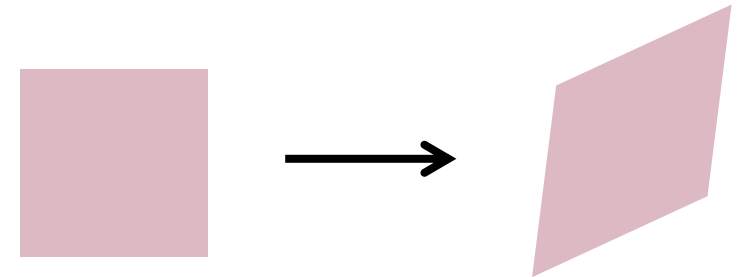
Affine transformations are combinations of

- arbitrary (4-DOF) linear transformations; and
- translations

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Properties of affine transformations:

- **origin does not necessarily map to origin**
- lines map to lines
- parallel lines map to parallel lines
- ratios are preserved
- compositions of affine transforms are also affine transforms



Does the last coordinate w ever change?

Affine transformations

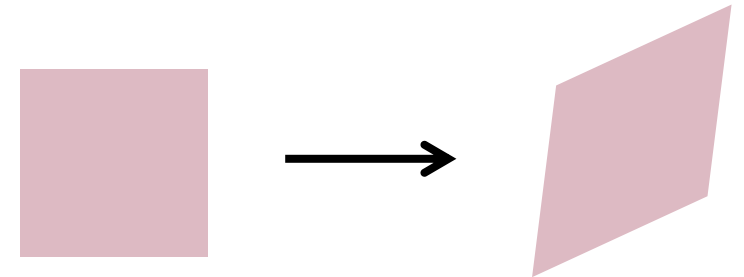
Affine transformations are combinations of

- arbitrary (4-DOF) linear transformations; and
- translations

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

Properties of affine transformations:

- **origin does not necessarily map to origin**
- lines map to lines
- parallel lines map to parallel lines
- ratios are preserved
- compositions of affine transforms are also affine transforms



Nope! But what does that mean?

How to interpret affine transformations here?

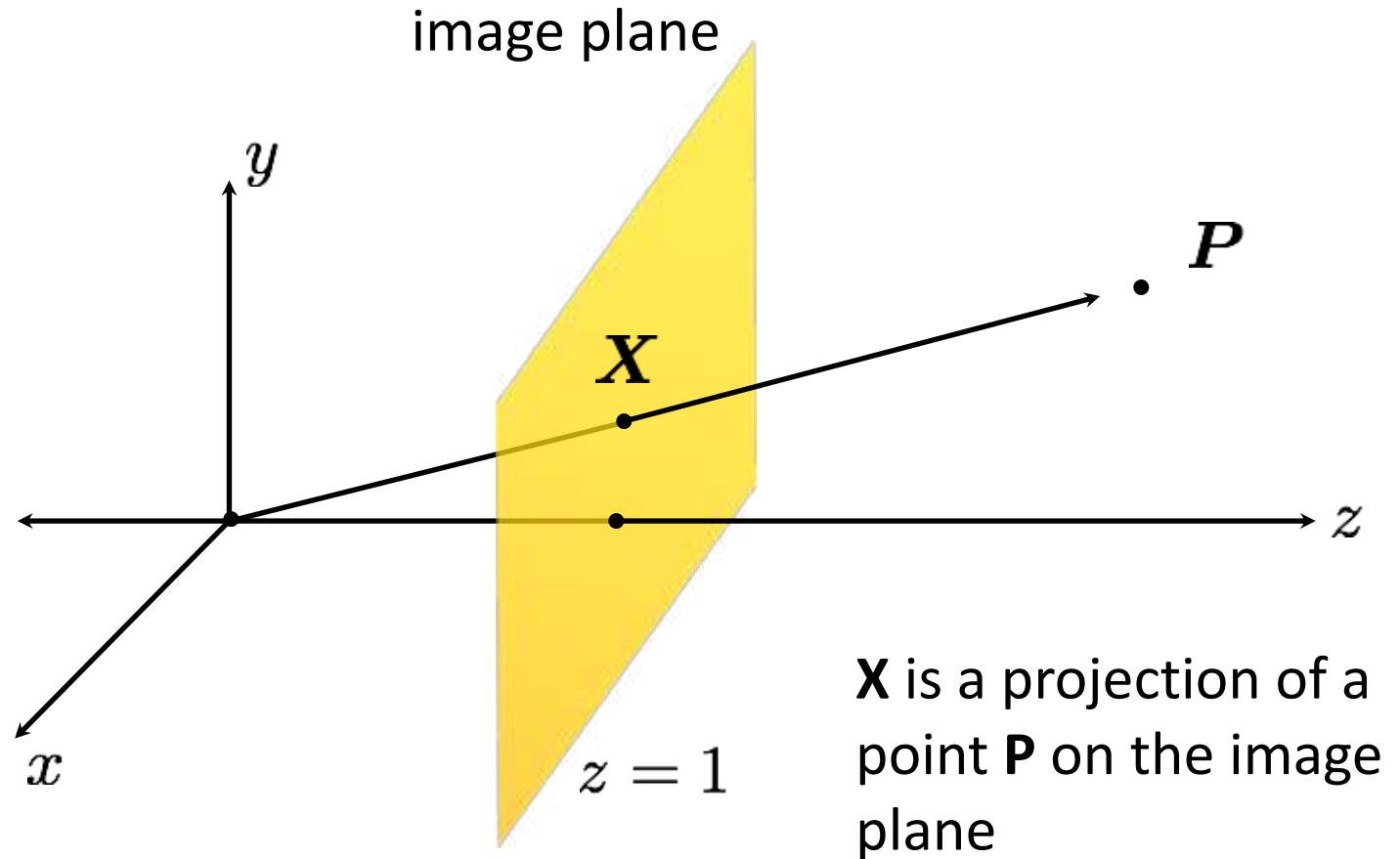
image point in
pixel coordinates

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$



image point in
heterogeneous
coordinates

$$\mathbf{X} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Projective transformations (aka homographies)

Projective transformations are combinations of

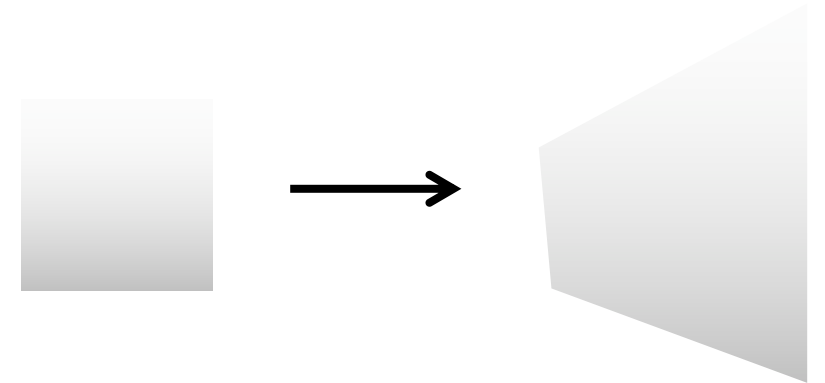
- affine transformations; and
- projective wraps

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

How many degrees of freedom?

Properties of projective transformations:

- origin does not necessarily map to origin
- lines map to lines
- parallel lines do not necessarily map to parallel lines
- ratios are not necessarily preserved
- compositions of projective transforms are also projective transforms



Projective transformations (aka homographies)

Projective transformations are combinations of

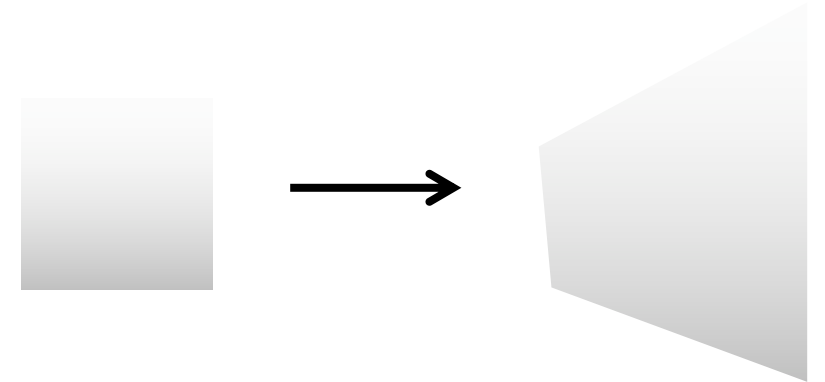
- affine transformations; and
- projective wraps

$$\begin{bmatrix} x' \\ y' \\ w' \end{bmatrix} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ w \end{bmatrix}$$

8 DOF: vectors (and therefore matrices) are defined up to scale)

Properties of projective transformations:

- origin does not necessarily map to origin
- lines map to lines
- parallel lines do not necessarily map to parallel lines
- ratios are not necessarily preserved
- compositions of projective transforms are also projective transforms



How to interpret projective transformations here?

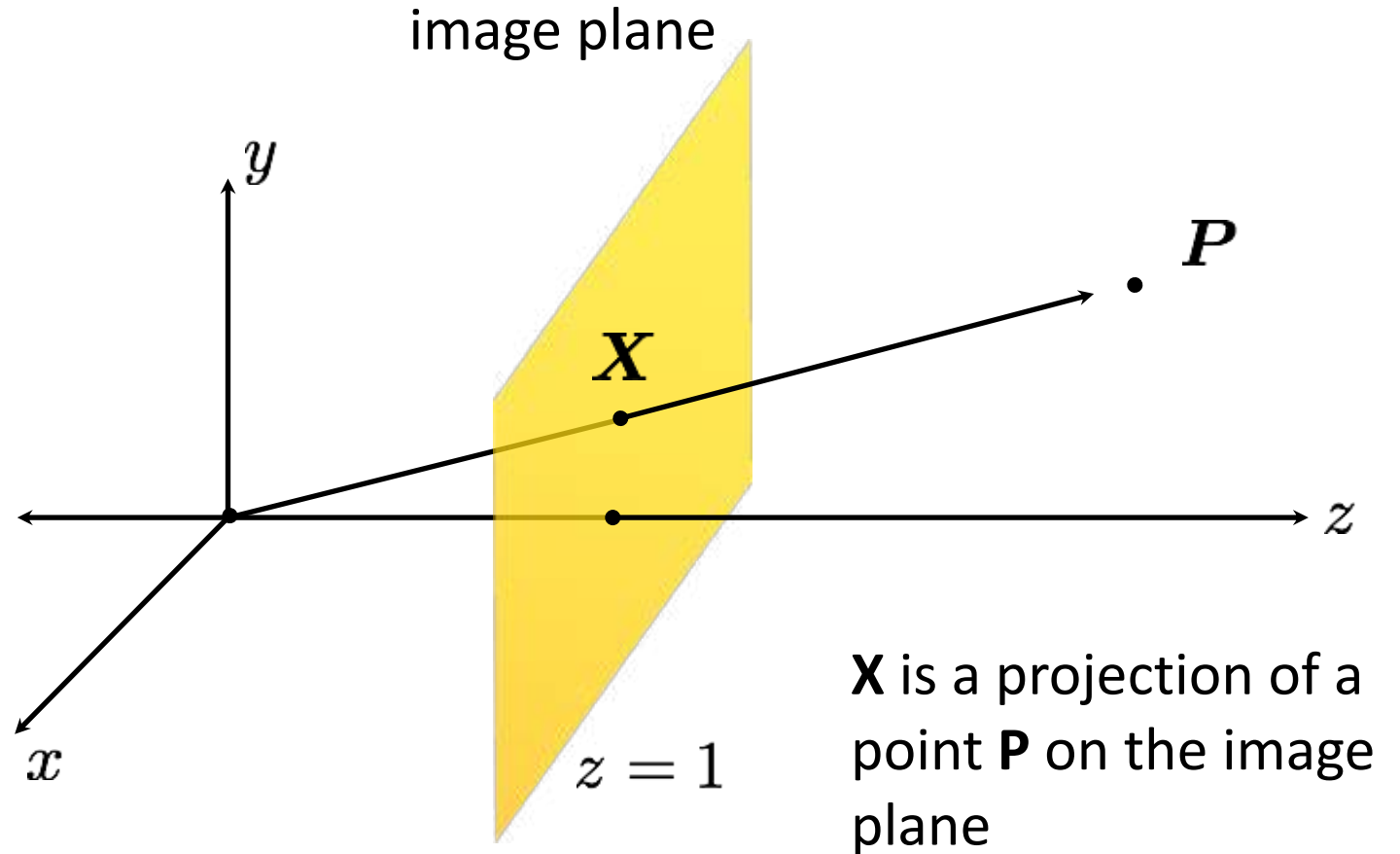
image point in
pixel coordinates

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

↕

image point in
heterogeneous
coordinates

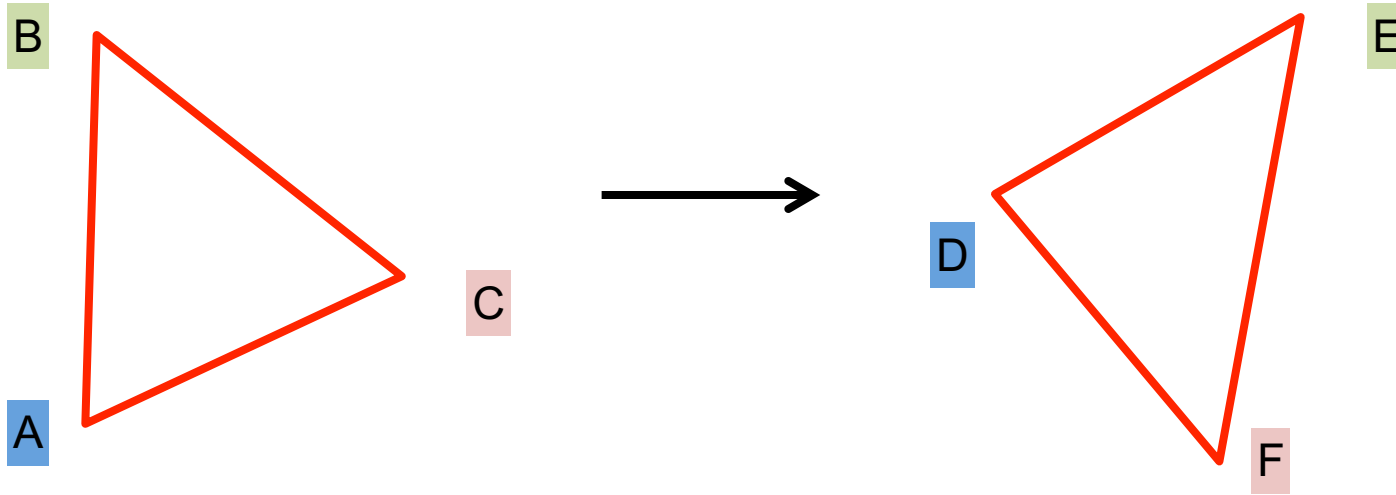
$$\mathbf{X} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Determining unknown (affine) 2D transformations

Determining unknown transformations

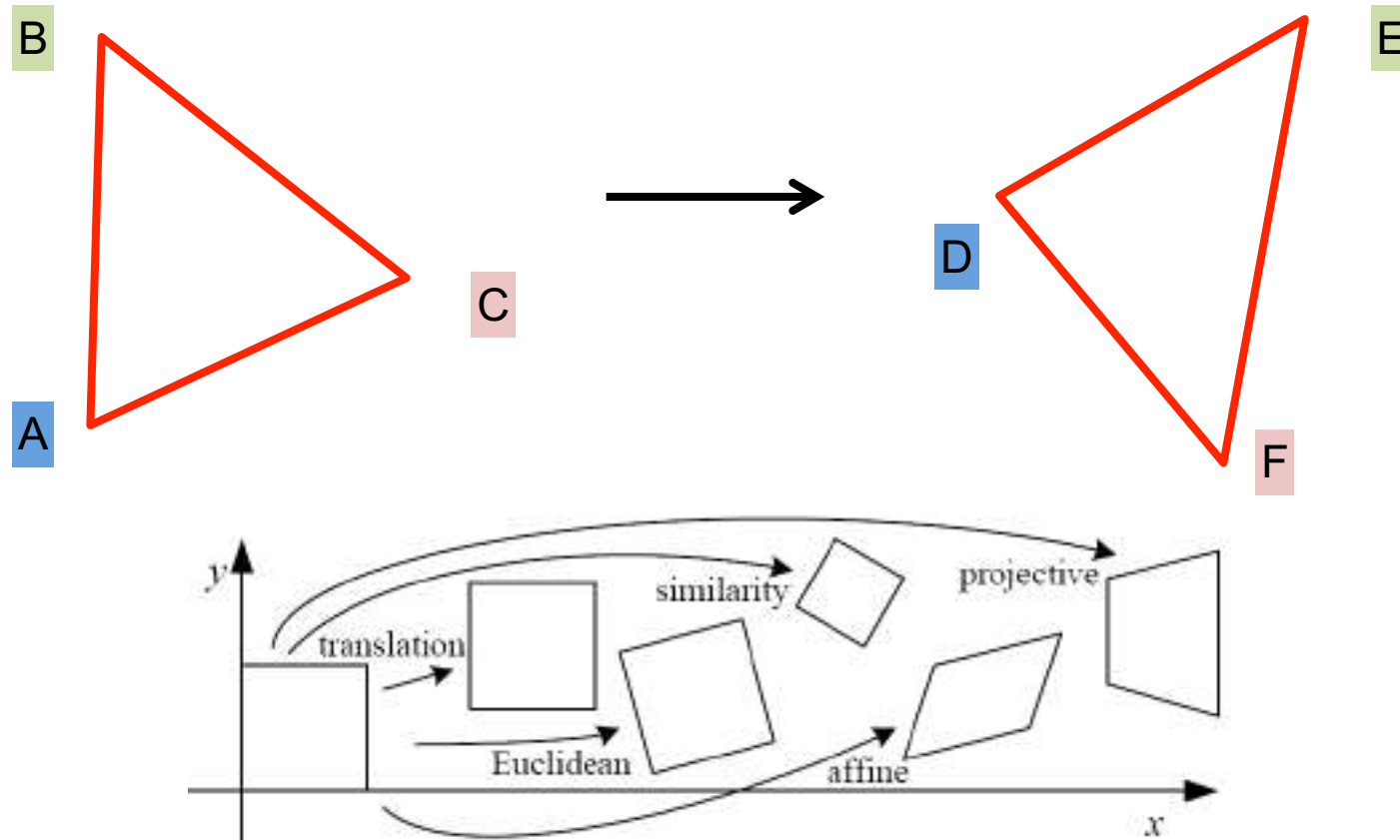
Suppose we have two triangles: ABC and DEF.



Determining unknown transformations

Suppose we have two triangles: ABC and DEF.

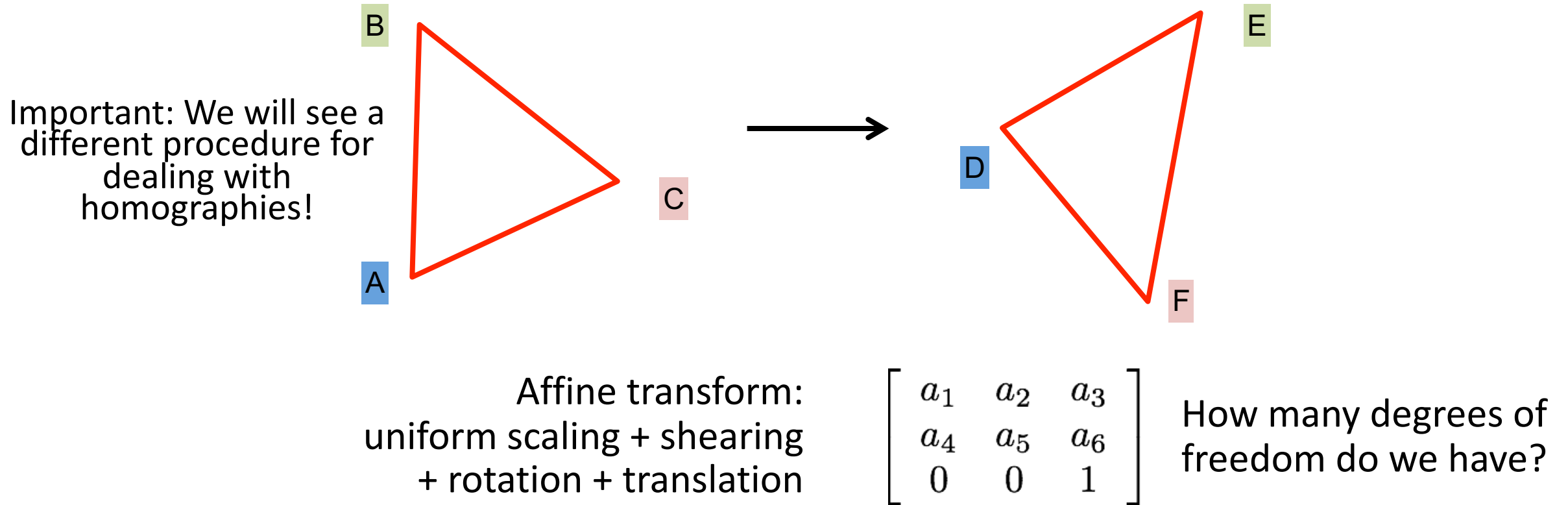
- What type of transformation will map A to D, B to E, and C to F?



Determining unknown transformations

Suppose we have two triangles: ABC and DEF.

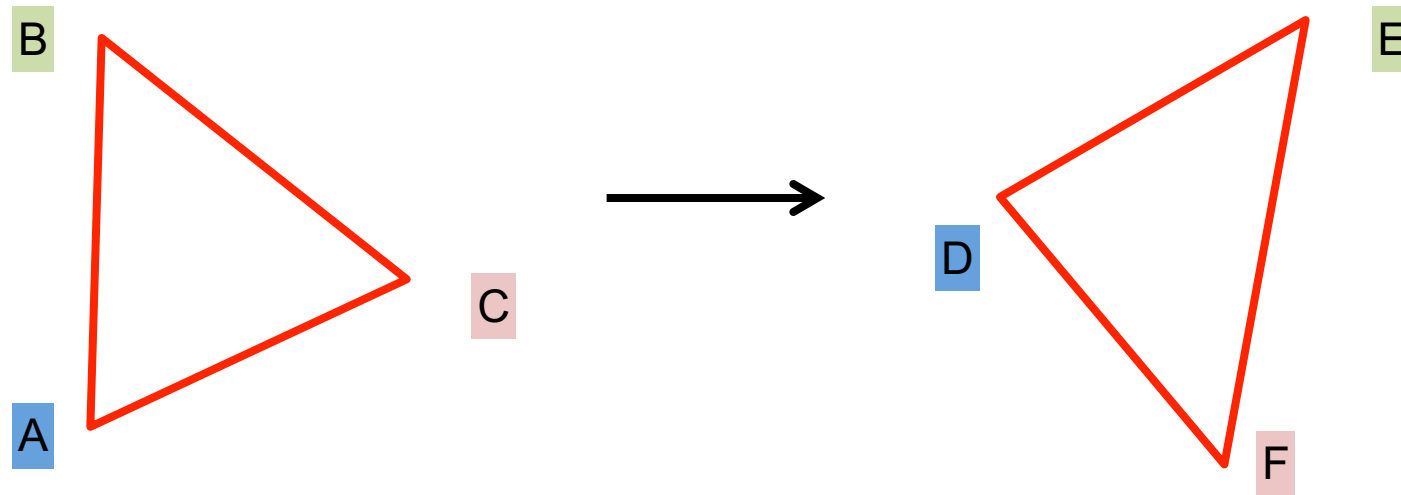
- What type of transformation will map A to D, B to E, and C to F?
- How do we determine the unknown parameters?



Determining unknown transformations

Suppose we have two triangles: ABC and DEF.

- What type of transformation will map A to D, B to E, and C to F?
- How do we determine the unknown parameters?



unknowns

$$x' = Mx$$

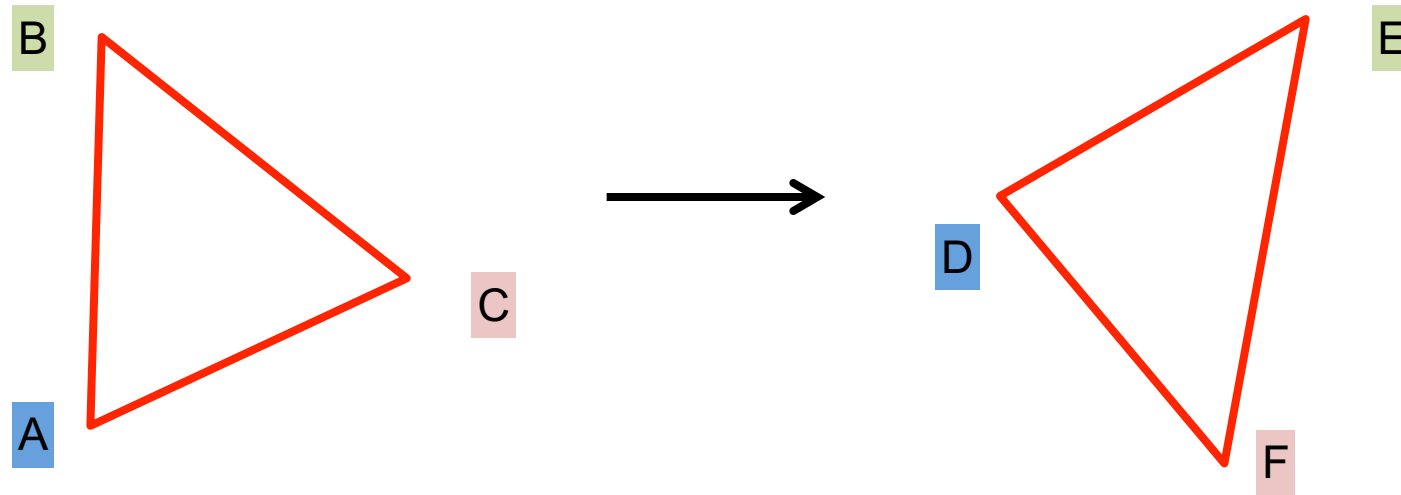
point correspondences

- One point correspondence gives how many equations?
- How many point correspondences do we need?

Determining unknown transformations

Suppose we have two triangles: ABC and DEF.

- What type of transformation will map A to D, B to E, and C to F?
- How do we determine the unknown parameters?

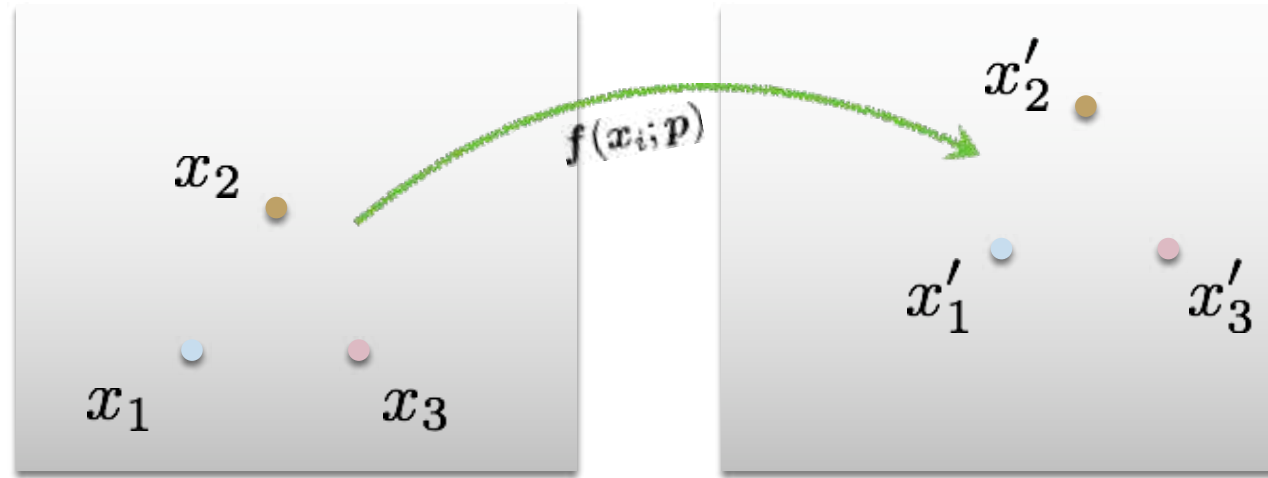


unknowns

$$x' = Mx$$

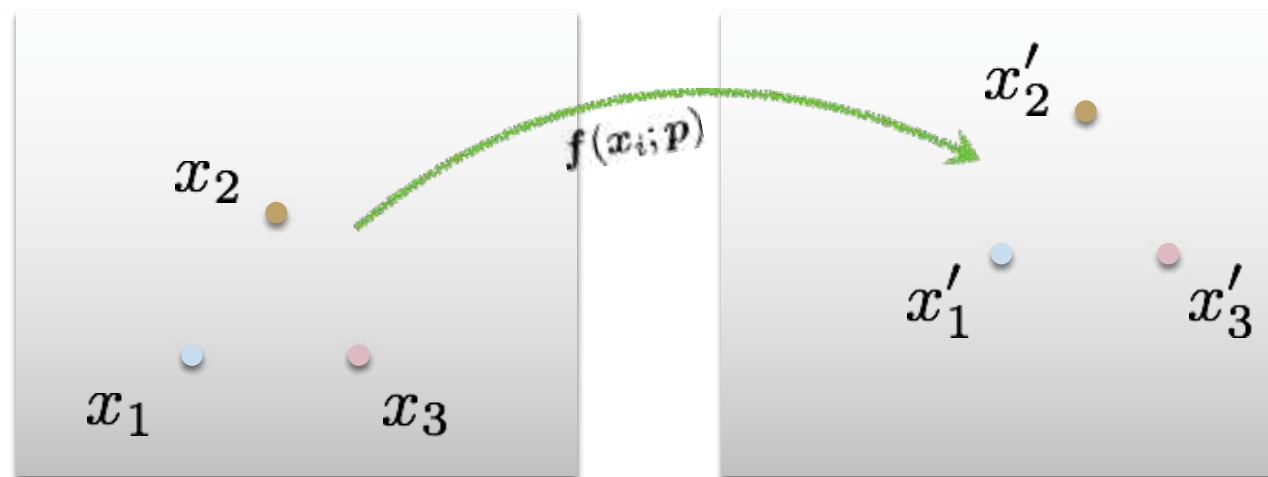
point correspondences

How do we solve this for **M**?



Least Squares Error

$$E_{\text{LS}} = \sum_i \|\mathbf{f}(\mathbf{x}_i; \mathbf{p}) - \mathbf{x}'_i\|^2$$



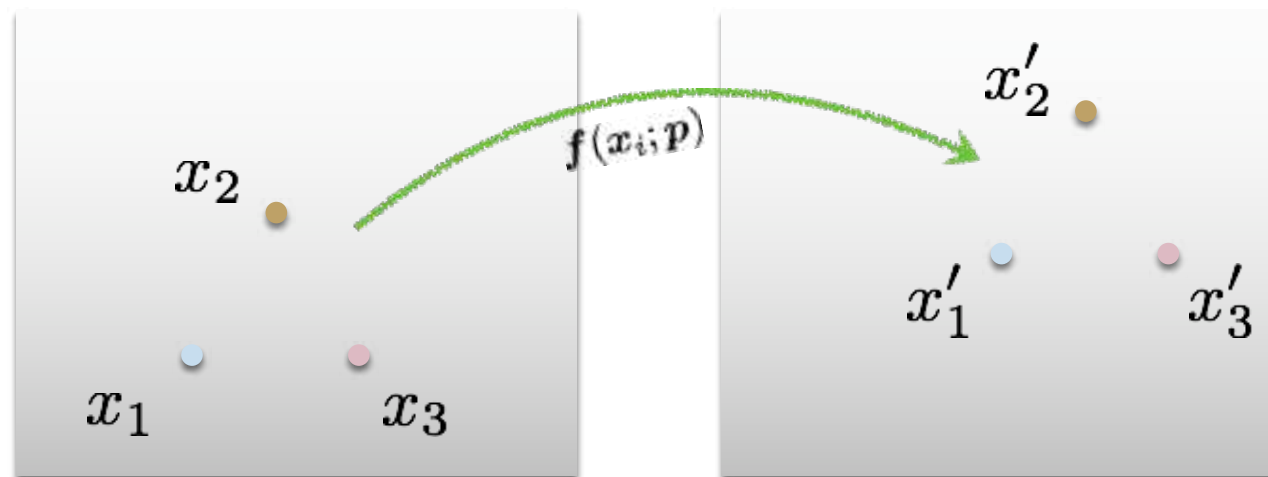
Least Squares Error

$$E_{\text{LS}} = \sum_i \left\| \mathbf{f}(x_i; \mathbf{p}) - x'_i \right\|^2$$

What is this?

What is this?

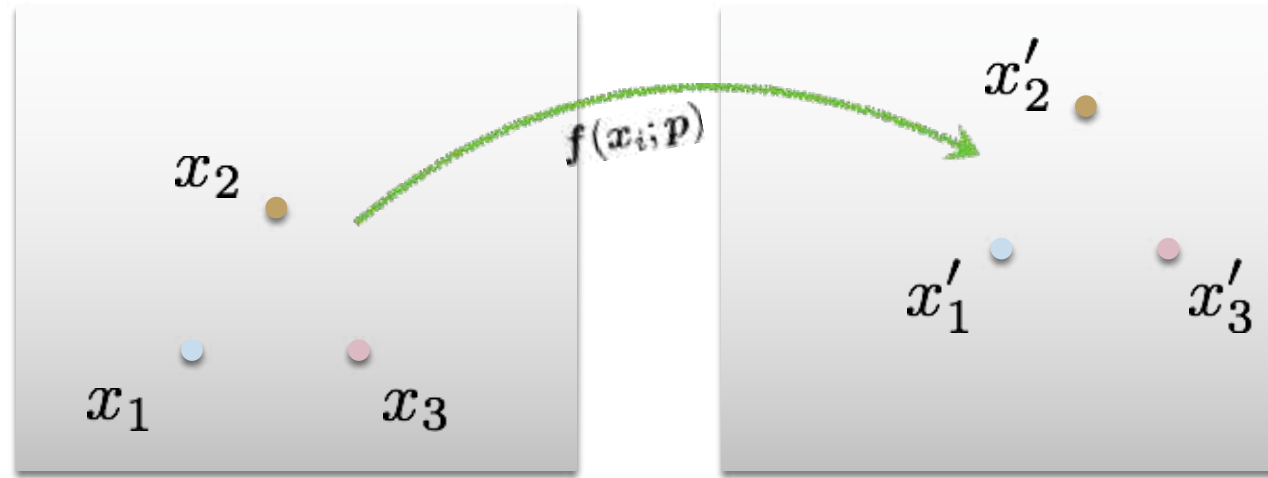
What is this?



$$\|x\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

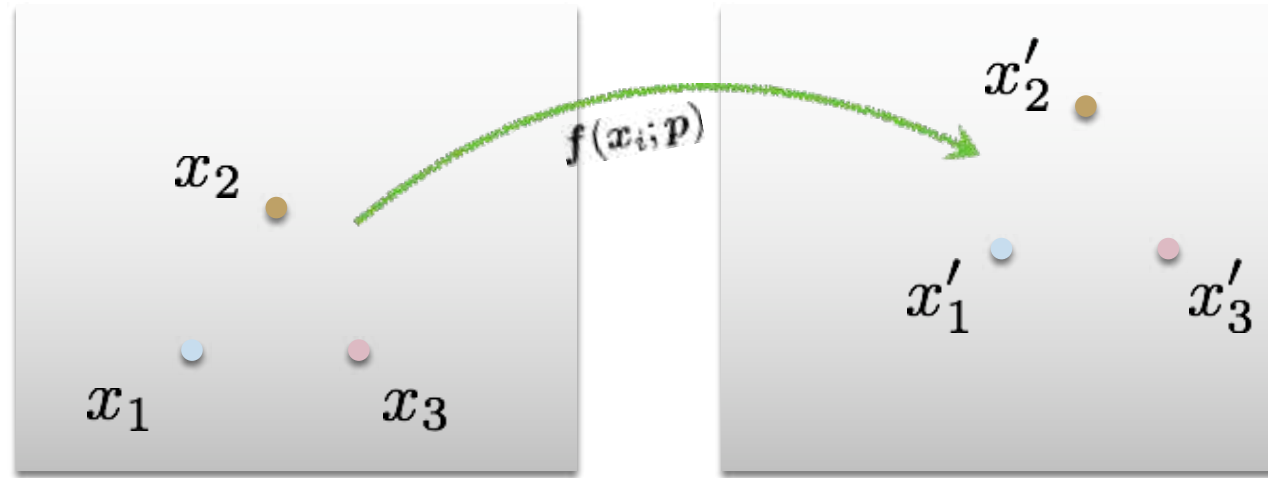
Least Squares Error

$$E_{\text{LS}} = \sum_i \left\| \underset{\substack{\uparrow \\ \text{predicted} \\ \text{location}}}{f(x_i; p)} - \underset{\substack{\uparrow \\ \text{measured} \\ \text{location}}}{x'_i} \right\| \overset{\substack{\uparrow \\ \text{Euclidean} \\ \text{(L2) norm}}}{\| \cdot \|} \overset{\text{squared!}}{^2}$$



Least Squares Error

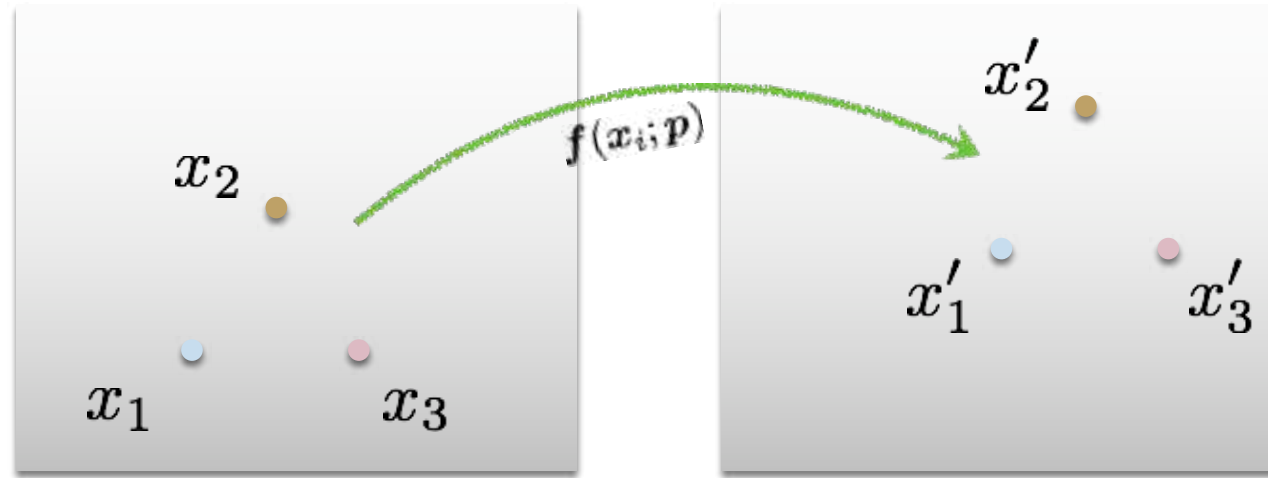
$$E_{\text{LS}} = \sum_i \underbrace{\| \mathbf{f}(\mathbf{x}_i; \mathbf{p}) - \mathbf{x}'_i \|^2}_{\text{Residual (projection error)}}$$



Least Squares Error

$$E_{\text{LS}} = \sum_i \|\mathbf{f}(\mathbf{x}_i; \mathbf{p}) - \mathbf{x}'_i\|^2$$

What do we want to optimize?



Find parameters that minimize squared error

$$\hat{\mathbf{p}} = \arg \min_{\mathbf{p}} \sum_i \|\mathbf{f}(\mathbf{x}_i; \mathbf{p}) - \mathbf{x}'_i\|^2$$

General form of linear least squares

(**Warning:** change of notation. $\mathbf{x}$ is a vector of parameters!)

$$\begin{aligned} E_{\text{LLS}} &= \sum_i |\mathbf{a}_i \mathbf{x} - \mathbf{b}_i|^2 \\ &= \|\mathbf{A} \mathbf{x} - \mathbf{b}\|^2 \quad (\text{matrix form}) \end{aligned}$$

Determining unknown transformations

Affine transformation:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} p_1 & p_2 & p_3 \\ p_4 & p_5 & p_6 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Why can we drop
the last line?

Vectorize transformation
parameters:

$$\begin{bmatrix} x' \\ y' \\ x' \\ y' \\ \vdots \\ x' \\ y' \end{bmatrix} = \begin{bmatrix} x & y & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x & y & 1 \\ x & y & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x & y & 1 \\ \vdots & & & \vdots & & \\ x & y & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x & y & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \\ p_6 \end{bmatrix}$$

Stack equations from point
correspondences:

$$\underbrace{\begin{bmatrix} x' \\ y' \end{bmatrix}}_{\mathbf{b}} = \underbrace{\begin{bmatrix} x & y & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x & y & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \\ p_5 \\ p_6 \end{bmatrix}}_{\mathbf{x}}$$

Notation in system form:

$\mathbf{b}$

$\mathbf{A}$

$\mathbf{x}$

General form of linear least squares

(**Warning:** change of notation. $\mathbf{x}$ is a vector of parameters!)

$$\begin{aligned} E_{\text{LLS}} &= \sum_i |\mathbf{a}_i \mathbf{x} - \mathbf{b}_i|^2 \\ &= \|\mathbf{A} \mathbf{x} - \mathbf{b}\|^2 \quad (\text{matrix form}) \end{aligned}$$

This function is quadratic.

How do you find the root of a quadratic?

Solving the linear system

Convert the system to a linear least-squares problem:

$$E_{\text{LLS}} = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

Expand the error:

$$E_{\text{LLS}} = \mathbf{x}^\top (\mathbf{A}^\top \mathbf{A}) \mathbf{x} - 2\mathbf{x}^\top (\mathbf{A}^\top \mathbf{b}) + \|\mathbf{b}\|^2$$

Minimize the error:

Set derivative to 0 $(\mathbf{A}^\top \mathbf{A})\mathbf{x} = \mathbf{A}^\top \mathbf{b}$

Solve for $\mathbf{x}$ $\mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b}$ ←

In Python:

```
x = numpy.linalg.  
solve(A, b)
```

Note: You almost never want to compute the inverse of a matrix.

Linear least squares estimation only works when the transform function is ?

Linear least squares estimation only works when the transform function is **linear! (duh)**

Also doesn't deal well with outliers